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Review article

Air-breathing polymer electrolyte fuel cells: A critical review of recent progress and future research needs

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ABSTRACT

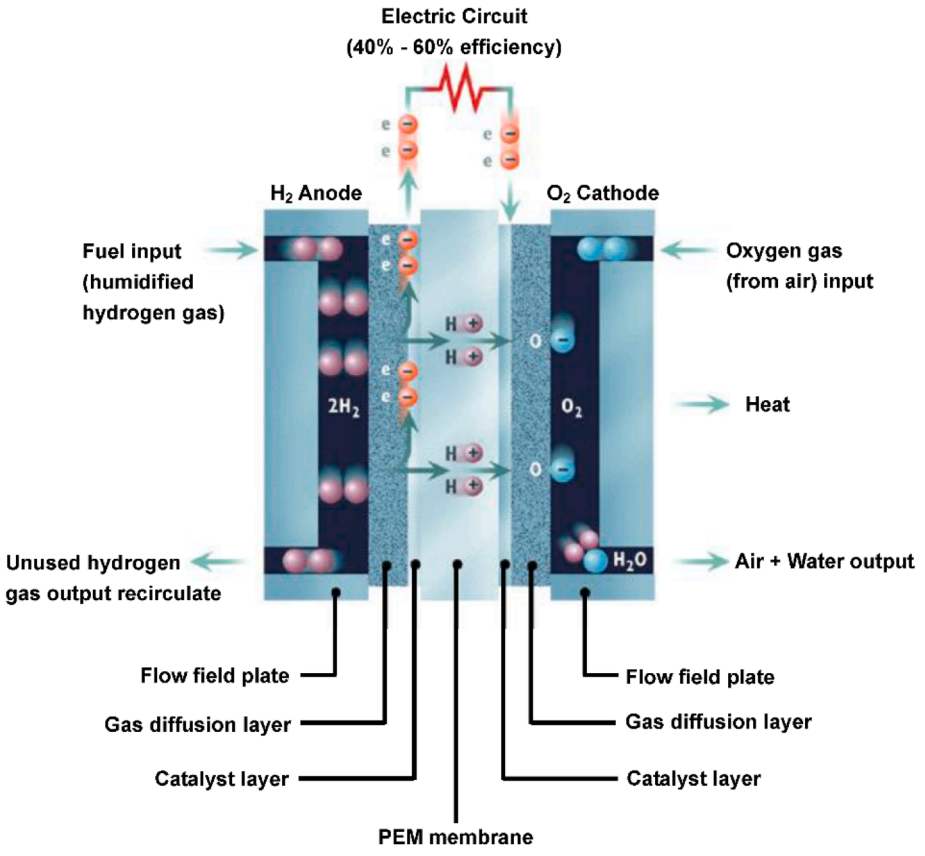
Air-breathing polymer electrolyte fuel cells (PEFCs) are a simplified alternative to conventional forced-convection PEFC systems, eliminating external air-supply devices to reduce system complexity, weight, and cost, and to minimise parasitic losses. Despite these advantages, their passive configuration poses critical challenges in oxygen mass transport, water management, thermal regulation, and environmental sensitivity. This study presents a comprehensive bibliometric and systematic review of air-breathing PEFC research published between 2000 and 2025. Using the Scopus database and a PRISMA-guided screening framework, 224 relevant documents were analysed through the Bibliometrix package in RStudio and VOSviewer to map publication trends, international collaborations, keyword evolution, and thematic research clusters. The results indicate irregular but sustained growth in research output, with an annual growth rate of 8.67%. Keyword co-occurrence analysis identified major research themes centred on water management, gas diffusion layer optimisation, flow-field design, numerical modelling, thermal diagnostics, and durability enhancement. Emerging niches include piezoelectric-assisted mass transport and advanced membrane materials for high-temperature and miniature systems. The findings highlight that future technological progress depends on improved mass-transport engineering, enhanced membrane stability, optimised passive flow architectures, and integrated stack-level thermal management. For researchers, the study clarifies the field's conceptual evolution and highlights the most critical research gaps. For industry stakeholders, it provides strategic insights into design priorities for compact, reliable hydrogen-powered systems suitable for portable, off-grid, and unmanned aerial vehicle applications.

1. Introduction

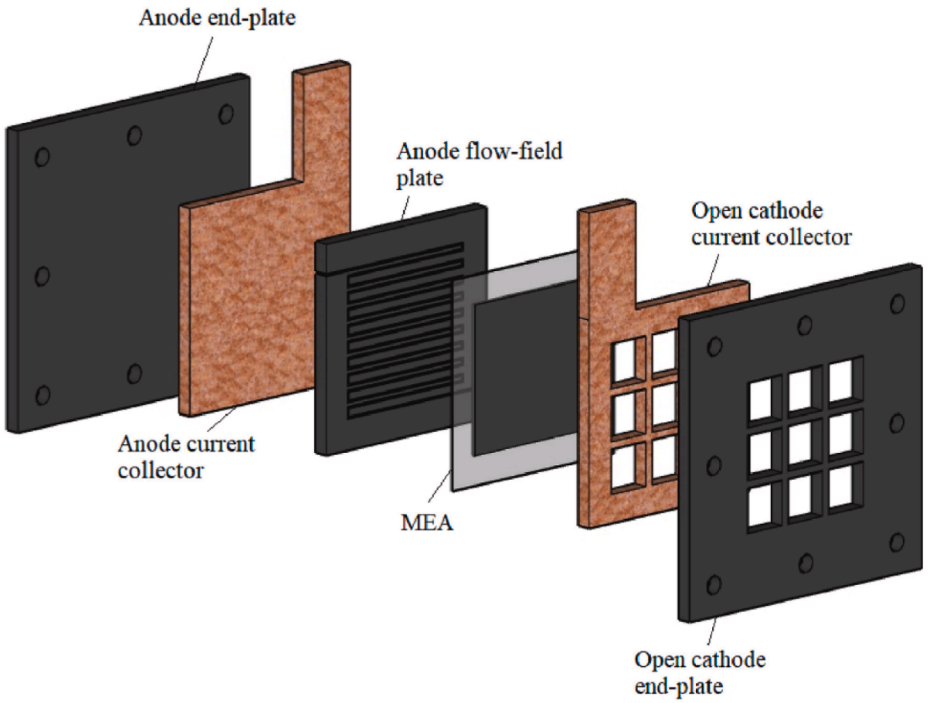
Globally, decarbonising the energy sector to address climate change has accelerated the transition to clean, efficient energy conversion technologies. In this context, hydrogen energy is a sustainable option for achieving carbon neutrality due to its zero-emission potential and versatility [1]. Accordingly, hydrogen fuel cell technologies are pivotal for achieving net-zero emissions, especially in sectors where direct electrification is insufficient [2–5]. In addition, global commitments under frameworks such as the United Nations Sustainable Development Goals (SDGs) have emphasised the need to develop low-emission power generation systems that are efficient, scalable, and adaptable to distributed applications. Among these, polymer electrolyte fuel cells (PEFCs) are a highly promising solution due to their high power density (>500 mW/cm^2), fast start-up capabilities, operating temperature of 60–80 °C, and an efficiency of about 60% [6]. PEFCs produce only water at the point of use, making them a zero-emission pathway for clean energy conversion [7]. Fig. 1(a) shows the typical operating principle of

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(a)



(b)

Fig. 1. (a) Typical operating principle of a PEFC [28] and (b) Exploded view of an air-breathing PEFC showing its key components. Reprinted from [10] with permission from Elsevier.

a PEFC. PEFCs function by electrochemically combining hydrogen and oxygen, with protons selectively transported across a polymer electrolyte membrane and electrons flowing through an external circuit to generate electricity. The core component, the membrane electrode assembly (MEA), facilitates these reactions and mass transfer processes [8,9].

Although conventional PEFCs provide numerous advantages for various applications, including portable devices, transportation, and stationary power generation, they typically rely on forced-convection air-supply systems that employ compressors or blowers [10–13]. This active air management introduces several complexities, such as increased system bulk, parasitic power losses, noise generation, and higher manufacturing costs [10]. Such limitations are evident in applications requiring miniaturisation, portability, or silent operation, such as small electronic devices, drones, and wearable energy systems [14–16].

To overcome these challenges, air-breathing polymer electrolyte fuel cells (air-breathing PEFCs), also referred to as passive air-cathode fuel cells, have attracted considerable attention [17]. In this context, passive refers to operation without externally powered air-supply systems, relying primarily on natural convection and diffusion for oxygen transport. Fig. 1(b) shows an exploded view of an air-breathing PEFC, showing its key components. This configuration eliminates the need for external air-supply systems by using natural convection and diffusion to deliver oxygen from ambient air directly to the cathode [16,18]. The natural simplicity of air-breathing designs, characterised by an open cathodic structure and minimal peripheral devices, reduces the balance-of-plant components, enhancing system reliability and reducing cost and weight [10,15,16,18].

Despite these benefits, the air-breathing configuration introduces several critical technical challenges that must be addressed for widespread adoption. These include managing oxygen mass transport limitations, ensuring uniform reactant distribution across the cathode, mitigating cathode flooding or drying, and addressing thermal management issues under passive conditions [10,19–21]. However, the performance and durability of air-breathing PEFCs are susceptible to environmental factors, including humidity, altitude, and temperature, making consistent operation difficult [22–24]. Water management, in particular, is a crucial operational challenge, requiring a delicate balance to prevent membrane dehydration and cathode flooding, which can severely impact performance and durability [20,25,26]. High current density operation further exacerbates these issues, leading to complex limitations in mass transfer, catalyst activity, and degradation [27].

However, recent innovations have focused on mitigating these challenges through advanced material science and engineering design. For instance, optimising key parameters such as current collector design and material, cathode gas diffusion layer (GDL) design and material, catalyst layer composition, and cell orientation has been shown to improve performance [10]. Other developments include biomimetic, integrated GDLs inspired by alveoli to enhance gas diffusion, water management, and thermal management [29]. The GDL, typically made of porous carbon, is pivotal in reactant distribution and water removal, with properties influenced by manufacturing processes and compression [30,31]. Likewise, novel polymer electrolyte membranes, including nanofiber-based membranes and those incorporating epoxidised natural rubber, are developed to enhance conductivity and stability [32–34]. Similarly, air-cooled multi-stack systems, often featuring micro-heat-pipe arrays, are explored to improve heat transfer efficiency and performance for higher-power applications [35,36].

These ongoing advancements are expanding the potential applications of air-breathing PEFCs to various flexible electronics, wearable devices, and off-grid micro-power units [14]. Foldable air-breathing fuel cells, for instance, are developed to operate effectively in various positions without an external air supply, addressing the limitations of conventional flexible fuel cells [18]. Though the commercialisation of PEFCs is scaling rapidly, efforts are still focused on overcoming manufacturing challenges and improving durability [37,38]. Therefore, continued research, leveraging tools such as X-ray computed tomography for water distribution analysis and machine learning for performance optimisation, is essential for advancing air-breathing PEFC technology towards broader commercial viability [39–42].

As shown in Table 1, previous reviews have addressed important technical aspects of PEMFC and air-breathing PEFC technologies. However, none of the reviewed studies provides a comparable integrated assessment of the research field itself. The existing reviews do not methodically quantify publication growth, country-level contributions, international collaboration, keyword co-occurrence, thematic structure, or thematic evolution. Therefore, the present study addresses this gap by combining bibliometric analysis and qualitative interpretation of the literature. Accordingly, this study presents an up-to-date analysis of research trends and innovations in air-breathing PEFCs through a combined quantitative bibliometric and qualitative systematic review framework.

To accomplish this, the review addresses the following specific objectives: (i) to analyse annual publication trends in the field; (ii) to identify leading countries and patterns of international collaboration; (iii) to examine dominant and emerging themes; (iv) to evaluate the evolution of the field's conceptual structure; and (v) to determine key research clusters through keyword co-occurrence analysis. In this way, the present study builds upon the technical knowledge established by previous reviews and addresses their broader analytical gap. Its main novelty is the integration of a technical review with quantitative research mapping, enabling the identification of established, emerging, and limited research themes within the overall air-breathing PEFCs literature. Therefore, this study provides a more comprehensive basis for understanding the current state of the field and defines future research needs.

The remainder of this review paper is structured as follows. Section 2 describes the methodological framework adopted for the study. Section 3 examines the current state of research and development in air-breathing PEFCs. Section 4 synthesises the key findings and identifies emerging research gaps and future directions. Finally, Section 5 presents the main conclusions of the review.

2. Methodology

Fig. 2 presents the methodological framework adopted in this study. A systematic bibliometric approach was employed, combining PRISMA-guided literature screening with quantitative bibliometric analysis to ensure transparency, reproducibility, and methodological rigour throughout the document identification and selection process [47,48]. It should be noted that integrating bibliometric analysis with a qualitative review signifies a significant contribution that distinguishes this work from existing narrative reviews.

2.1. Literature search and screening process

The Scopus database was selected as the sole source for data extraction, consistent with other notable publications [51–55]. Scopus is also one of the most comprehensive multidisciplinary abstract and citation databases available. The database search was conducted on February 4, 2026, using the following search string: TITLE-ABS-KEY ("air-breathing" OR "air breathing") AND ("polymer electrolyte fuel cell" OR "PEM fuel cell" OR "PEMFC"), which targeted the title, abstract, and keywords fields of indexed documents to maximise retrieval sensitivity and maintain thematic specificity. The search was restricted to publications from 2000 to 2025 to capture the modern development of air-breathing PEFC research. The search initially identified 251 documents across all Scopus subject areas. The dataset was then refined using predefined inclusion and exclusion criteria. First, only documents indexed in subject areas relevant to the research scope were retained, namely Energy, Engineering, Chemistry, Physics and Astronomy,

Table 1
Comparison of previous PEMFC and air-breathing PEFC reviews and the research gap addressed.

Authors	Objective	Contributions	Major limitation	Gap addressed by this study
Calili-Cankir et al. [10]	To review how key design and operating parameters influence the performance of air-breathing PEFCs, including current collectors, cathode GDLs, catalyst layers, cell orientation, ambient conditions, hydrogen storage/delivery, and stack/system development.	Provides a focused assessment of the major cell components and operating conditions that affect AB-PEFC performance and discusses challenges to wider commercialisation.	Primarily provides a technical/component-oriented review and does not systematically map publication trends, geographical contributions, collaboration networks, or the evolution of research themes.	Provides a broader research-domain perspective by identifying publication trends, country contributions, collaboration patterns, dominant research themes, thematic evolution, and relationships among research topics.
Sapkota et al. [15]	To review recent developments in planar PEMFCs, specifically air-breathing planar configurations, with emphasis on their potential for thin, lightweight, flexible, and portable applications.	Highlights material, fabrication, durability, stack design, and cost-reduction requirements for planar and flexible PEMFCs.	Focuses mainly on planar PEMFC design and portable applications, rather than providing a comprehensive assessment of the air-breathing PEFCs research field.	Examines the overall development and evolution of air-breathing PEFCs research, including themes that extend beyond planar configurations and individual design considerations.
Ji & Wei [28]	To comprehensively review water-management processes in PEMFCs, especially the balance between membrane hydration and water flooding, together with their characterisation, modelling, effects, and mitigation.	Provides a detailed foundation for understanding water-management mechanisms, flooding, degradation, and mitigation strategies in PEMFCs.	Focuses mainly on water management in PEMFCs generally and does not address the broader research domain or the distinctive research directions of air-breathing PEFCs.	Places water management within the broader and more recent air-breathing PEFCs research domain and identifies how this and other research themes have evolved.
Hwang & Sung [43]	To review recent PEMFC design developments aimed at achieving lightweight and flexible configurations, including tubular structures, materials, manufacturing processes, and performance-related design factors.	Demonstrates structural and material approaches for reducing PEMFC weight and improving flexibility while maintaining performance.	Concentrates on structural design, lightweight construction, and flexibility, without mapping the broader air-breathing PEFCs research domain.	Identifies the broader research structure, emerging themes, and the evolution of themes in air-breathing PEFC studies rather than focusing on a particular structural configuration.
Huang et al. [44]	To critically review passive water and thermal management in air-breathing PEFCs and assess their importance for performance, durability, affordability, and commercialisation.	Integrates water and thermal management considerations and highlights the need to couple these processes to improve air-breathing PEFC performance and durability.	Although directly concerned with air-breathing PEFCs, its objective is centred on water and thermal management, rather than the complete research landscape.	Expands the perspective from water/thermal management to the full spectrum of air-breathing PEFCs research, including research growth, thematic clusters, emerging topics, and future research directions.
Bunyan et al. [45]	To evaluate the effects of flow-field configurations and geometric parameters on PEMFC performance, especially reactant distribution, pressure drop, water management, and mass transport.	Synthesises the advantages and limitations of conventional, nature-inspired, and modified flow-field configurations.	Focuses on flow-field design parameters and does not establish the knowledge, thematic, or collaborative structure of air-breathing PEFCs research.	Moves beyond individual flow-field parameters to identify dominant, emerging, and interconnected research themes across the air-breathing PEFCs literature.
Punyawudho et al. [46]	To review advances in GDL, catalyst-layer, and polymer-electrolyte materials and assess their influence on PEMFC efficiency, cost, durability, and performance.	Provides detailed coverage of electrode and electrolyte materials and identifies material optimisation strategies for improving PEMFC performance and durability.	Primarily adopts a materials/component perspective and considers PEMFCs broadly rather than systematically assessing the AB-PEFC research field.	Integrates material-related research into a field-wide analysis of air-breathing PEFCs research, and identifies research trends, thematic relationships, emerging areas, and future needs.

Materials Science, Chemical Engineering, Environmental Science, and Multidisciplinary fields. This resulted in 250 documents. Second, the document type was restricted to peer-reviewed journal articles and conference papers, while book chapters, reviews, editorials, notes, and other document types were excluded, resulting in 238 documents. Third, only English-language publications were retained to ensure consistency in interpretation and analysis, producing a final pre-screening dataset of 229 documents.

Following data extraction, the titles and abstracts of all 229 documents were independently screened by two reviewers to assess eligibility and thematic relevance to the study's objectives. This dual-reviewer approach was adopted to minimise subjectivity and reduce the risk of inclusion bias during the screening phase. During this process, 5 documents were identified outside the defined scope of this review and were then excluded. Disagreements between the two reviewers during the screening process were resolved through discussion and consensus to ensure the integrity of the final selection. This resulted in a final dataset of 224 documents for the bibliometric analyses. The 224 documents, retained in CSV format, were subsequently imported into the Bibliometrix package in RStudio and VOSviewer.

2.2. *Bibliometric analysis*

The Bibliometrix package, an open-source tool developed for science mapping and bibliometric analysis in the R programming environment [56,57], was employed to conduct a set of quantitative analyses including: (i) publication trend analysis through annual article production, which provided insights into the growth trends of research activity in the field; (ii) country-level research contribution and international collaboration network analysis, which mapped the geographic contributions of scholarly output and the nature of cross-national research partnerships; (iii) keyword frequency and co-occurrence analysis, which identified the most prominent and recurring thematic terms within the field; and (iv) conceptual structure analysis comprising thematic mapping, thematic evolution, and factorial analysis of keywords, allowing a dynamic understanding of how research themes have emerged, matured, or declined over the study period. Complementing these analyses, VOSviewer, a widely used software tool for constructing and visualising bibliometric networks [58,59] was employed to generate network visualisation maps of keyword co-occurrence, which facilitated the identification and clustering of related research themes. These keyword clusters were not interpreted in isolation but were justified and contextualised through a qualitative review of existing studies sourced from the broader literature. This approach ensured that the quantitative

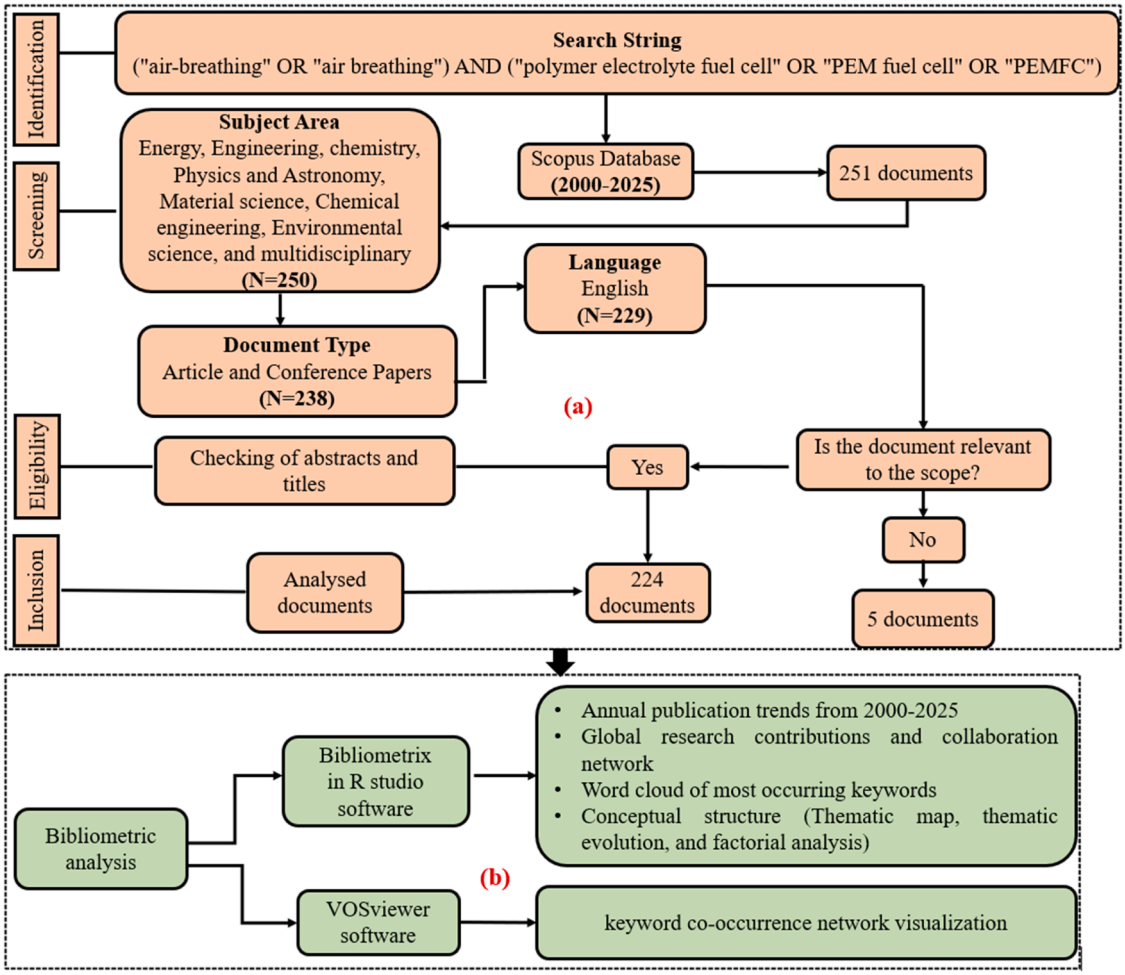


Fig. 2. (a) PRISMA-based flow diagram of the document selection process and (b) Bibliometric analysis workflow using Bibliometrix (RStudio) and VOSviewer. Adapted and Modified from [49,50].

bibliometric findings were grounded in substantive domain knowledge and aligned with established theoretical and technological narratives within the air-breathing PEFC research domain.

3. Results and discussion

This section presents and discusses the results generated using RStudio-based Bibliometrix and VOSviewer software. It also provides a comprehensive analysis of the key findings and their broader implications.

3.1. Overview of bibliometric parameters

Fig. 3 summarises the key bibliometric indicators in the dataset. The dataset comprises 224 documents across 94 sources, with an annual growth rate of 8.67%. This growth trend indicates sustained interest in



Fig. 3. Summary of bibliometric indicators on air-breathing PEFCs research.

air-breathing PEFC technology. This could be driven by increased demand for portable power applications and miniaturised fuel cell systems. The figure shows a collaborative research domain field contributed by 706 authors with an average of 9.55 co-authors per document. The absence of single-authored documents underlines the interdisciplinary and collaborative nature of fuel cell research, which requires expertise in electrochemistry, materials science, and engineering. However, the international co-authorship rate of 21.88% suggests that research efforts remain mostly confined to national or institutional boundaries, potentially limiting knowledge exchange and the diffusion of innovation across geographic regions.

Similarly, the figure shows that the average number of citations per document is 23.59, and the average document age is 12.8 years. The citation rate suggests reasonable visibility and influence, though it may indicate that air-breathing PEFCs occupy a niche within the broader fuel cell sector. These findings highlight opportunities for enhanced international collaboration and suggest that, although the field continues to grow, strategic efforts to foster cross-border partnerships could accelerate technological advancement and the commercial viability of air-breathing fuel cell systems.

3.2. Trend in yearly publications in air-breathing polymer electrolyte fuel cells research

Fig. 4 displays the number of articles published annually over the 25 years. It can be observed that the early phase (2000–2004) produced minimal research output, with only irregular publications. This indicates

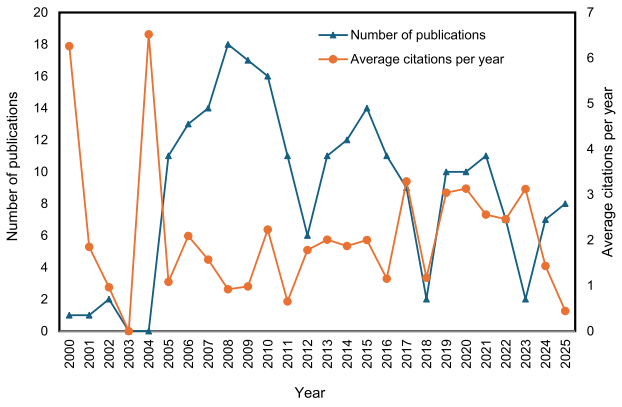


Fig. 4. Annual document publications on air-breathing PEFCs research.

the technology's early developmental stage and limited contribution to the research community. However, publications increased to 11 articles in 2005, marking the beginning of an intensive research period that persisted through 2011. This increase in total publications coincides with several factors, including global energy security concerns following oil price volatility, the implementation of the Kyoto Protocol's first commitment period (2005–2012) [60], and governmental investments in hydrogen and fuel cell technologies. Notably, the United States Department of Energy's Hydrogen Programme [61] and similar initiatives in Japan, South Korea, and European nations could have provided research funding during this period. The peak productivity in 2008 (18 articles) aligned with climate change discourse and pre-financial-crisis optimism regarding clean energy technologies [62], demonstrating that policy momentum and funding availability correlate with research output.

However, research publications declined to 6 articles in 2012. This could be attributed to the delayed impact of the 2008–2009 global financial crisis on research budgets. The observed stabilisation in research publications during 2019–2021 (10–11 articles) could be attributed to renewed climate commitments under the Paris Agreement and interest in hydrogen-economy concepts. However, the subsequent decline in 2022–2023 suggests continued challenges in maintaining research drive. The slight recovery in 2024–2025 (7–8 articles) potentially indicates emerging niche applications or integration with broader hydrogen infrastructure initiatives.

Insights from the top 30 most cited paperstions. However, citation counts tend to favour older publications, while recent high-impact studies could not yet have had sufficient time to accumulate citations. In addition, citation patterns can be influenced by journal visibility, reputation, and established research networks. Therefore, citation counts should be interpreted with caution and not treated as a direct measure of research quality or scientific impact. The studies in the table span a broad range, from membrane chemistry and electrode design to thermal management and system-level integration, advancing the understanding of passive fuel cell technology for portable, lightweight power applications. Among the most foundational contributions to this field is the work of Moore et al. [63], published in the Journal of Power Sources and receiving 169 total citations, which examined the susceptibility of PEFCs to battlefield contaminants introduced into the oxidant air stream. The results showed that common pollutants, such as sulphur dioxide, nitrogen dioxide, and benzene, caused negligible or reversible performance degradation. In contrast, chemical warfare agents, including sarin, sulphur mustard, cyanogen chloride, and hydrogen cyanide, compromised fuel cell performance in a wholly irreversible manner, raising significant concerns about the deployment of air-breathing cells in contaminated environments.

Building on advances in membrane science, Moghaddam et al. [64] introduced a silicon-based inorganic-organic PEM as an alternative to Nafion. This study has received 148 citations and was published in Nature Nanotechnology. The authors revealed that by fabricating pores of about 5–7 nm in diameter within a silicon substrate and functionalising the pore surfaces with a self-assembled molecular monolayer capped with porous silica, the resulting membrane achieved proton conductivity two to three orders of magnitude higher than that of Nafion under low-humidity conditions. It delivered an order-of-magnitude greater power density with a dry hydrogen feed and an air-breathing cathode. Santarosa et al. [65], whose work has received 140 citations in the International Journal of Hydrogen Energy, investigated an eight-cell open-cathode PEFC stack in which a dual-function air fan served as both an oxidant supply and a thermal management device. The results showed that the stack achieved a maximum power density of 310 mW/cm^2 at 790 mA/cm^2 under ambient conditions (temperature and pressure). However, excessive temperature increases led to membrane dehydration and an increase in ohmic resistance. This study demonstrates the essential balance between thermal and hydration management in forced-convection air-breathing stacks. The work of Fabian et al. [66], cited 116 times in the Journal of Power Sources, systematically varied ambient temperature and relative humidity to evaluate a planar air-breathing cell on dry hydrogen in dead-end mode. Using shadowgraphy and impedance spectroscopy, the authors identified three operational regimes: partial membrane hydration with increasing current density, full hydration with GDL flooding, and membrane dry-out. It was seen that the transition from full hydration to dry-out occurred at a GDL surface temperature of about 60 °C, irrespective of ambient conditions, with a maximum power density of 356 mW/cm^2 achieved at 20 °C and 40% relative humidity.

Similarly, Jeong et al. [22], whose paper has received 114 citations in the Journal of Power Sources, investigated the effects of cathode open area and ambient relative humidity on the performance of passive air-breathing PEFCs. The authors found that a cathode open area of 77% yielded the highest performance among the configurations tested, ranging from 52% to 94%. In addition, increasing ambient relative humidity improved performance in the low-current-density region. However, it caused flooding-related losses at higher current densities, suggesting a balance between oxygen accessibility and membrane hydration. Ning et al. [67], published in ACS Nano and with 104 total citations, developed a flexible, lightweight air-breathing PEFC using a novel composite electrode design. The results showed that a 1×1 cm^2 cell weighing as little as 0.065 g and measuring just 0.22 mm in thickness achieved a specific volume power density of 5190 W/L, retained 89.1% of its original performance after 600 bending cycles, and withstood repeated drops from 30 m This represented a significant step toward fuel cells suitable for flexible electronics and wearable devices. Bussayajarn et al. [68], work published in the International Journal of Hydrogen Energy with 91 citations, proposed planar air-breathing PEFCs without endplates and evaluated three cathode designs (parallel slit, circular open, and oblique slit) at the same opening ratio using self-humidifying membrane electrode assemblies. The results show that circular opening geometry yielded the best performance due to its shortest rib distance and smallest hydraulic diameter. However, fuel cell instability was observed under both self-breathing and forced convection conditions due to water accumulation that could not be removed solely by natural evaporation. González-Espasandín et al. [69], published in Renewable Energy with 89 citations, modelled the performance of air-breathing hydrogen PEFCs and passive direct methanol fuel cells under simulated unmanned aerial vehicle (UAV) atmospheric flight conditions. The results showed that low pressure at altitude affected PEFC/hydrogen performance more severely than direct methanol fuel cell (DMFC) performance. At the same time, the atmospheric relative humidity influenced PEFC performance at higher cell temperatures, while DMFC performance remained comparatively stable. Both cell types demonstrated the feasibility of low- to medium-altitude UAV propulsion with appropriate thermal controls.

Likewise, Jiang et al. [70], in a paper published in the Journal of

Materials Chemistry with 89 citations, highlighted that sodium dodecylbenzenesulfonate-adsorbed holey graphene oxide membranes are effective proton exchange membranes for air-breathing DMFCs. The membranes exhibited proton conductivity comparable to that of Nafion 112, along with substantially lower methanol permeability, thereby delivering higher performance and greater stability than Nafion-based counterparts. Ying et al. [71], in a study published in the Journal of Power Sources with 87 total citations, developed a validated three-dimensional computational model of an air-breathing PEFC that incorporated free convection, electrochemical reactions, and water transport. The study concluded that cathode channel width influenced performance and that an open ratio of approximately 75.9% yielded optimal results. Similarly, Meyer et al. [72] conducted a novel simultaneous mapping of current density and temperature in an air-cooled open-cathode PEFC stack. It was observed that at higher current densities, segments exceeding 70 °C experienced membrane dehydration and local current limiting, while cooler segments compensated by carrying higher current. Dead-ended anode operation generated spatio-temporal gradients that required about 840 s to reach steady state before full recovery upon reopening the purge valve. Schmitz et al. [73] analysed the influence of cathode opening size and GDL wetting properties. The authors found that the largest opening ratio (80%) combined with an untreated GDL produced the best performance at 100 mW/cm^2 and 43 °C, and that neither strongly hydrophilic nor highly hydrophobic diffusion layers improved upon the slightly hydrophobic baseline material. Paquin and Fréchette [21] offered a comprehensive analytical and experimental framework for understanding flooding and dry-out in small air-breathing PEFCs. The authors established that the ratio of thermal to mass transport resistance determines whether a cell operates in the flooding, stable, or dry-out regime, providing a design framework for balancing thermal rejection and oxygen transport. In a separate contribution on materials innovation, Wei et al. [74] studied carbon nanotube-based GDL's found that a three-step fabrication process- pore creation, microporous layer coating, and low-temperature heat treatment at 350 °C produced a GDL <40 μm thick, achieving a volume-specific power density of 15,600 W/L in an air-breathing PEFC.

Also, Xiong et al. [75] introduced a dual-catalyst-layer cathode structure designed to improve water management, with an inner hydrophilic Nafion-bound layer maintaining proton transfer efficiency and an outer hydrophobic layer expelling excess water at high current densities. The results showed greatly improved air-breathing performance through optimised gas-diffusion and microporous-layer parameters. Ying et al. [76] developed a coupled three-dimensional model for air-breathing PEFCs validated against temperature distribution and polarisation data. The authors revealed that concentration losses and natural convection are the primary determinants of cell performance and that ambient relative humidity influences operation. Fernández--Moreno et al. [77] demonstrated a complete portable power system based on a single air-breathing PEFC fed by a metal hydride hydrogen cartridge. The study showed that anodic purging and cathode fan assistance improved steady-state response by facilitating the removal of excess water, and that the system could deliver >1 W of DC power for 20 hrs using just 1 g of hydrogen. Baroutaji et al. [78] explored the use of open-pore cellular foam as a flow distributor in an air-breathing PEFC. It was found that polytetrafluoroethylene (PTFE) coating of the metallic foam enhanced hydrophobicity and corrosion resistance, and that the PTFE-coated design outperformed uncoated foam configurations. However, the air-breathing design still exhibited poorer water management than pressurised-air configurations. Fabian et al. [79] addressed cathode flooding under severe conditions by integrating a hydrophilic porous graphite wicking layer at the cathode, with the results showing stable operation across all tested current densities from 100 to 600 mA/cm^2 for six continuous hours under flooding-prone conditions of 10 °C and 80% relative humidity, compared to complete voltage collapse in the un-wicked control cell. Ying et al. [80] extended Nguyen's model to incorporate buoyancy-driven natural convection and concentration overpotential, demonstrating that concentration losses are more severe under natural convection than under forced convection in narrow channels. That performance cannot be improved indefinitely by widening channels, necessitating cathode flow-field optimisation.

Correspondingly, in the work of Litster et al. [81], the authors conducted a two-dimensional computational analysis of micro-structured cathodic electrodes with nanoporous gas-diffusion media. The study found that Knudsen diffusion within nanoporous structures does not impose insurmountable mass-transfer limitations and that the electrode configuration supports passive temperature and oxygen regulation through natural convection above the horizontal cathode surface. Lee et al. [82] developed a cylindrical air-breathing PEFC with a helical flow channel. Their numerical analysis and pressure-sensitive film measurements revealed that the cylindrical configuration converted clamping pressure into a uniformly distributed contact pressure, yielding a maximum power density of 220 mW/cm^2, about 24% higher than a comparable planar design. In the work of Hojung et al. [83], the authors investigated the effect of hydrophilic silica nanoparticles in the anode catalyst layer. They found that 40 wt% SiO_2-doped membrane electrode assemblies improved air-breathing performance by 44.4% compared to undoped assemblies, attributing the improvement to enhanced anode water management under dry hydrogen conditions. Ismail et al. [84] constructed an efficient mathematical model for air-breathing PEFCs. They demonstrated that neglecting Joule heating or entropic heat sources overestimates cell performance, and that assuming liquid-phase water production considerably alters Nernst potential predictions, with optimal ambient conditions shifting depending on the cell operating potential. Wang and Ouyang [85] developed a coupled three-dimensional computational fluid dynamics (CFD) model using non-dimensional heat and mass transfer coefficients. They proposed a simplified correlation relating the Sherwood number to the sum of the Grashof and diffusional Grashof numbers, providing a useful tool for understanding oxygen transport limitations and guiding optimal air-breathing PEFC design.

Furthermore, Xu et al. [86] experimentally and numerically investigated cathode flow field configurations in an air-breathing high--temperature PEFC. It was observed that the opening shape and orientation exerted less influence than the open ratio, with slot holes at a 75% open ratio delivering a maximum power density of 160 mW/cm^2 at 394 mA/cm^2. Obeisun et al. [87] evaluated printed circuit board flow-field plates with parallel and circular geometries at varying opening ratios. The authors found that circular openings reduced ohmic resistance but increased mass-transport limitations. In contrast, greater opening ratios improved mass transport at the expense of increased contact resistance, with the best design achieving 250 mW/cm^2. Dun et al. [88] developed Fe-N-doped hierarchical mesoporous carbon catalysts as platinum alternatives for the oxygen reduction reaction, demonstrating a half-wave potential only 32 mV more negative than that of commercial Pt/C and achieving approximately 63% of the Pt/C peak power density in an air-breathing PEFC at room temperature. Li et al. [89] characterised an air-breathing direct ethanol fuel cell with an anion-exchange membrane and platinum-free electrodes, achieving a peak power density of 38 mW/cm^2 at room temperature, comparable to that of conventional Nafion-based direct methanol fuel cells. However, cathode water flooding was identified as a limiting factor. Rajani and Kolar [90] presented a two-dimensional, steady-state, non-isothermal model for a vertical, planar air-breathing PEFC. The study showed that maximum power density increased with decreasing cathode height, decreasing ambient temperature, and increasing relative humidity, while warning that higher local temperatures in short cells risked exceeding safe operating limits at high current densities. Yan et al. [91] reported that combining cathode flow channel configurations with 50% and 58.3% opening ratios within an air-breathing stack reduced temperature and improved distribution uniformity. They also found that assisted-air-breathing conditions enhanced stack performance, with simulation and experimental data showing reasonable agreement when

average cross-sectional temperatures were used as the fitting parameter. Table 2

3.3. Most occurring keywords

Fig. 5 shows the word cloud of the most frequently used keywords in research on air-breathing PEFCs. The prevalence of words such as air-breathing, PEFC, fuel cells, air-breathing PEFC, water management, gas diffusion layer, hydrogen, natural convection, performance, CFD, membrane electrode assembly, and stack indicates that air-breathing polymer electrolyte fuel cells are an important technology. Air-breathing polymer electrolyte fuel cells use air instead of an air supply system, relying on natural convection to supply oxygen from ambient air. As such, words such as water management and gas diffusion layer

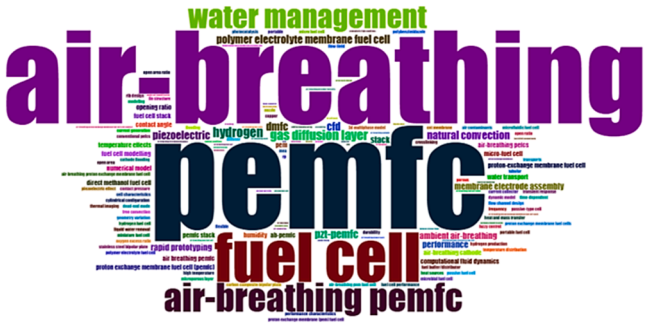


Fig. 5. Word cloud illustrating the most frequently used author keywords in air-breathing PEFC research.

Table 2
Top 30 most highly cited publications on air-breathing PEFC research.

Authors	Year	Main objective/ Contribution	Study type	Journal	Total Citations
Moore et al. [63]	2010	Investigated the effects of contaminants (pollutants and chemical agents) on PEFC performance	Experimental	Journal of Power Sources	169
Moghaddam et al. [64]	2010	Developed an inorganic-organic membrane with improved conductivity and performance	Experimental	Nature technology	148
Santarosa et al. [65]	2007	Examined operating conditions (temperature, airflow, pressure) on PEFC stack performance	Experimental	International Journal of Hydrogen Energy	140
Fabian et al. [66]	2006	Studied the influence of ambient temperature and humidity on air-breathing PEFC	Experimental	Journal of Power Sources	116
Jeong et al. [22]	2006	Analysed the effect of cathode open area and humidity on performance	Experimental	Journal of Power Sources	114
Ning et al. [67]	2017	Designed a flexible, lightweight air-breathing PEFC with high power density	Experimental	ACS Nano	104
Bussayajarn et al. [68]	2009	Evaluated different cathode geometries and self-humidifying MEAs	Experimental	International Journal of Hydrogen Energy	91
González-Espasandín et al. [69]	2019	Modeled PEFC/DMFC performance under unmanned Aerial Vehicles flight conditions	Modelling	Renewable energy	89
Jiang et al. [70]	2014	Developed a graphene oxide membrane for improved DMFC performance	Experimental	Journal of Materials Chemistry A	89
Ying et al. [71]	2005	Developed a 3D model to study channel configuration effects	Modelling	Journal of Power Sources	87
Meyer et al. [72]	2015	Mapped the current and temperature distribution in open-cathode PEFC	Experimental	Journal of Power Sources	85
Schmitz et al. [73]	2006	Investigated cathode opening size and gas diffusion layer properties	Experimental	Journal of Power Sources	68
Paquin & Fréchette [21]	2008	Studied flooding and dry-out via experiments and an analytical framework	Experimental	Journal of Power Sources	67
Wei et al. [74]	2020	Developed a novel carbon nanotube-based gas diffusion layer	Experimental	Journal of Materials Chemistry A	65
Xiong et al. [75]	2015	Improved water management using a dual catalyst layer design	Experimental	International Journal of Hydrogen Energy	64
Ying et al. [76]	2005	Combined simulation and experiment for PEFC performance analysis	Modelling/ Experimental	Journal of Power Sources	64
Fernández-Moreno [77]	2013	Developed a portable PEFC power system	Experimental	Applied Energy	63
Baroutaji et al. [78]	2017	Designed PEFC using open-pore cellular foam	Experimental	International Journal of Hydrogen Energy	60
Fabian et al. [79]	2010	Proposed passive water management using a wicking layer	Experimental	Journal of Power Sources	60
Ying et al. [80]	2005	Modelled the effect of cathode channel dimensions	Modelling	International Journal of Hydrogen Energy	60
Lister et al. [81]	2006	2D-Computational analysis of heat/mass transfer in the cathode	Modelling	Journal of Power Sources	55
Lee et al. [82]	2010	Developed cylindrical PEFC configuration	Experimental	International Journal of Hydrogen Energy	49
Hojung et al. [83]	2007	Improved water management using silica nanoparticles	Experimental	International Journal of Hydrogen Energy	49
Ismail et al. [84]	2014	Developed a mathematical model including heat effects	Modelling	Applied Energy	48
Wang & Ouyang [85]	2007	3D heat and mass transfer modelling in PEFC	Modelling	Journal of Power Sources	48
Xu et al. [86]	2019	Combined experimental and numerical study of flow field designs	Modelling/ experimental	International Journal of Hydrogen Energy	46
Obeisun et al. [87]	2014	Investigated printed circuit board-based flow-field geometries	Experimental	International Journal of Hydrogen Energy	46
Dun et al. [88]	2019	Developed Fe-based catalysts for the oxygen reduction reaction	Experimental	Journal of Mateials. Chemistry A	44
Li et al. [89]	20,113	Studied the performance of the ethanol fuel cell system	Experimental	International Journal of Hydrogen Energy	44
Rajani & Kolar [90]	2007	Developed a 2D model for vertical PEFC	Modeling	Journal of Power Sources	43
Yan et al. [91]	2020	Studied thermal management in PEFC stacks	Experimental/ modelling	International Journal of Hydrogen Energy	40

are important in air-breathing polymer electrolyte fuel cells. The presence of words such as CFD and temperature effects indicates that researchers are simulating air-breathing PEFCs to optimise performance [44,92]. The prevalence of words such as membrane electrode assembly also indicates that air-breathing polymer electrolyte fuel cells have moved beyond conceptual design to performance optimisation. These are important words that form part of the air-breathing polymer electrolyte fuel cells technology.

Furthermore, the appearance of terms such as CFD and temperature effects may indicate that modelling and simulation are being used extensively to optimise flow channels, heat transfer, and reactant distribution. The use of terms such as MEA and stack-level studies also indicates that the research has evolved from conceptual designs to performance optimisation. These themes are essential for engineering and represent basic engineering needs for improvement. Conversely, the less common keywords, including opening ratio, contact angle, high temperature, durability, micro-fuel cell, portable fuel cell, humidity, and air circulation, may reflect more specialised or developing areas of research. Also, variables such as opening ratio and contact angle are associated with micro-structural and surface engineering optimisation efforts that focus on improving passive airflow and water removal, concerns that have only arisen as the community transitions from development to performance optimisation. Durability and high-temperature performance are becoming more prominent as the community transitions from proof-of-concept designs to commercialisation, but these topics are often addressed through long-term testing and the use of advanced materials, which can be limited in scope. Micro-fuel cells and portable fuel cells also reflect application-oriented research for consumer electronics and remote power, which is a developing area of research but not yet mainstream hydrogen-fed PEFC [93,94]. The presence of direct methanol fuel cells in less frequently used keywords could be due to the development of separate but parallel research streams. In summary, the keyword usage in the word cloud indicates that the field is founded on fundamental electrochemical and transport problems but is gradually moving toward material innovation, durability, miniaturisation, and application-oriented optimisation, i.e., traits of technological development and expansion.

3.4. Keyword cluster analysis

Fig. 6 displays the keyword co-occurrence network visualisation generated using VOSviewer. This figure shows the major research themes and thematic relationships within the air-breathing PEFC literature. It is worth noting that the node size represents the frequency of keyword occurrence. At the same time, the link strength indicates the degree of co-occurrence between terms. The following subsections present 12 thematic clusters derived from bibliometric keyword grouping using VOSviewer. The node colours show the different clusters in the network. Each cluster is listed with its constituent keywords/terms, followed by a discussion of the research themes, implications, and significance represented by that grouping.

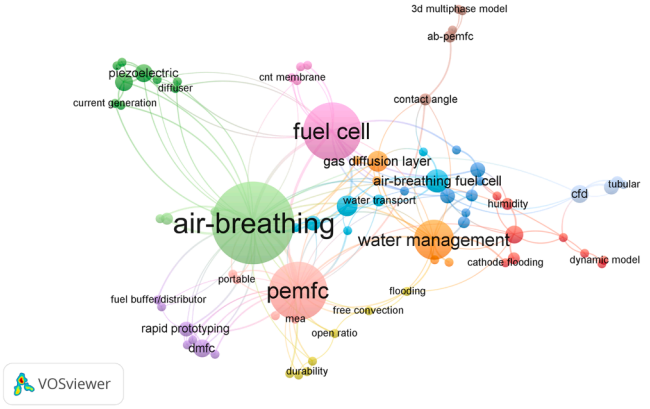


Fig. 6. Keyword co-occurrence network clustering research themes in air-breathing PEFC studies.

3.4.1. Thermal and mass transport phenomena in conventional PEFCs

Cluster 1 comprises the following terms: cathode flooding, conventional polymer electrolyte fuel cell (PEFC), dynamic model, humidity, hydrogen fuel cell, natural convection, transient response, and heat and mass transfer. This cluster centres on the fundamental thermo-fluid dynamics governing conventional PEFC operation. The cluster focuses on heat and mass transfer phenomena and their transient behaviour. The grouping of these keywords suggests a research orientation toward understanding coupled interactions. These interactions concern temperature, moisture, and airflow in hydrogen fuel cells operating under passive or semi-passive conditions. The term cathode flooding is one of the most critical operational challenges in air-breathing PEFCs. Liquid water, a by-product of the electrochemical cathode reaction, can cause flooding if not removed effectively. This flooding blocks the GDL pores and reduces oxygen access to the catalyst layer. In view of this, Liu et al. [95] developed a one-dimensional two-phase cathode model to address this. The study found that flooding in the catalyst layer far exceeds that in the GDL. In addition, liquid water saturation is governed by physical, material, and operating parameters. Therefore, removing water from the catalyst layer first is essential for preventing flooding. Park et al. [96] engineered crack-structured microporous layers to address this water removal challenge. The connected-crack GDL architecture created continuous liquid-water highways from the catalyst to the backing layers. This effectively separated liquid and gas transport pathways within the electrode. Three-dimensional X-ray tomography confirmed a significant reduction in flooding at the catalyst-microporous layer (MPL) interface. The crack-engineered GDL achieved a 20% higher peak power density of 1.23 W/cm^2. Pore-scale simulations further revealed a 25-fold enhancement in water removal efficiency.

Similarly, the inclusion of humidity alongside cathode flooding in this cluster emphasises the sensitivity of water balance to ambient conditions. This sensitivity is especially significant in open-cathode, air-breathing configurations without forced humidification. Kwon and Eom [97] demonstrated that oversaturated cathode conditions at 150% relative humidity induced severe MEA degradation. Cathode flooding triggered back-diffusion, causing collateral anode degradation through repeated flooding events. The authors proposed low-humidity electrochemical impedance spectroscopy as a diagnostic tool for early fault identification. Trinh et al. [98] further showed that cathode relative humidity above 50% increased oxygen transport resistance. This flooding-induced resistance reduced both limiting current density and maximum power density. At 80% relative humidity, limiting current density remained constrained at 1.563 A/cm^2 regardless of load acceleration time.

Likewise, “natural convection” indicates a key transport mechanism within this cluster's thematic framework. Unlike forced-convection systems that employ blowers, air-breathing fuel cells rely on buoyancy-driven airflow to supply oxygen. Natural convection acts as a thermal management tool and a reactant delivery mechanism. Calili-Cankir et al. [92] constructed parametric mathematical models comparing air-breathing and conventional PEFCs, as shown in Fig. 7. It was observed that conventional PEFCs outperformed air-breathing configurations. This performance gap stemmed from higher forced-convection heat- and mass-transfer coefficients. Conversely, the air-breathing PEFC operated under naturally convective, lower-coefficient conditions. Notably, air-breathing cells performed better as GDL porosity decreased and GDL thickness increased. Kim et al. [99] investigated ionomer optimisation in self-humidifying air--breathing PEFCs without external humidification (Fig. 8). The results confirmed the suitability of passive-type cells as compact mobile power sources, including drones.

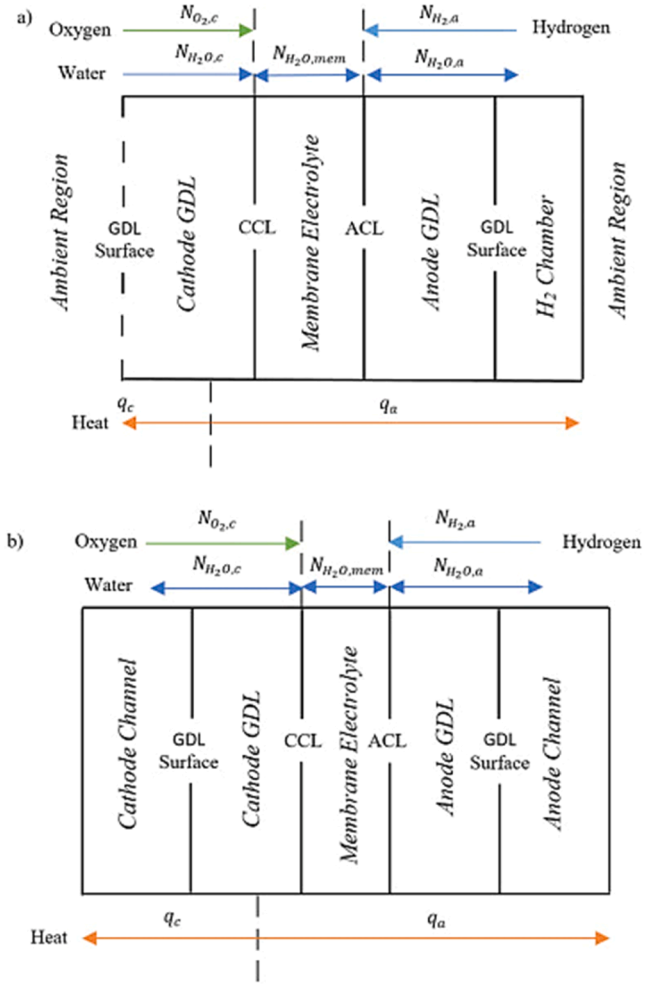


Fig. 7. Illustrative schematics of the modelled (a) air-breathing and (b) conventional PEFCs. Here, 'N' represents molar flux, 'q' denotes heat flux, 'CCL' is the cathode catalyst layer, and 'ACL' is the anode catalyst layer, with the subscripts 'a and 'c' referring to the anode and cathode, respectively [92]. Published under open access.

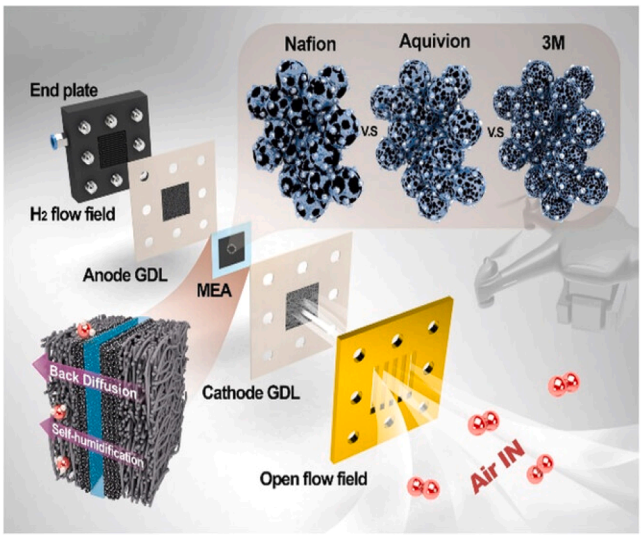


Fig. 8. Ionomer distribution at catalyst agglomerates in self-humidifying air-breathing PEFCs, comparing Nafion, Aquivion, and 3 M ionomers within the MEAs [99]. Published under open access.

The terms dynamic modelling and transient response are central to this cluster's research significance. Fuel cell operation involves frequent load changes, start-up cycles, and varying ambient conditions. These events induce transient behaviour that steady-state models cannot capture. Calili-Cankir et al. [100] developed comparative dynamic models for both air-breathing and conventional PEFCs. The study found the air-breathing PEFC's dynamic response to load alterations was significantly slower. Lower natural convection heat transfer coefficients caused poor heat dissipation and higher cell temperatures. This resulted in higher ohmic and activation losses during transient operation. Liu et al. [101] analysed the dynamic responses of startups and identified a "two-stage" behaviour under dry conditions. Under wet conditions, a more complex "three-stage" response mode with greater voltage fluctuation emerged. A step-loading strategy with gradually decreasing magnitude achieved a startup time of only 9.6 s. Abdi et al. [102] addressed performance under flooding conditions through multi-objective optimisation using genetic algorithms. The optimised operating parameters yielded a 38.94% increase in average current density. Simultaneously, the non-uniformity in current density decreased by 38.8%, improving both power output and distribution. Fault detection and tolerant control under flooding conditions are other significant research perspectives. To address this, Lv et al. [103] proposed a fuzzy-interval-observer-based fault-tolerant control method for cathode flooding. The Takagi-Sugeno fuzzy model reproduced the flooding conditions for fault detection. Chen et al. [104] developed an active fault-tolerant controller integrated with a water transfer model. The flooding fault indicator accurately identified flooding in real time, allowing effective recovery from performance issues. Experimental validation confirmed the controller's ability to reliably suppress water management failures.

Finally, it can be noted that this cluster addressed issues related to the dynamic behaviour of the thermofluid and the electrochemistry involved in the operation of the PEFC, especially with regard to cathode flooding, humidity effects, natural convection, and transients during passive operation of the cell. Studies have shown that dynamic modelling, crack engineering of the GDL/MPL, humidity control, and fault-tolerant controls can greatly enhance fuel cell performance. It was found that stable and self-regulating heat and water management during transient operation under passive natural-convection conditions remains a primary unresolved engineering limitation, as it presents challenges such as flooding, membrane dehydration, and uneven reactant transport that are difficult to control simultaneously.

3.4.2. Piezoelectric actuation for enhanced mass transport and water management in PEFCs

Cluster 2 contains terms including current generation, diffuser, frequency, nozzle, open area, piezoelectric (PZT), PZT-PEFC, and rib design. This cluster suggests an innovative, emerging niche within air-breathing PEFC research. It centres on the active integration of piezoelectric transducers to enhance mass transport and water management. The acronym PZT-PEFC captures this approach precisely, using vibrating piezoelectric elements to generate localised oscillating air-flows. Ma and Huang [105] introduced this concept through a three-dimensional transitional model of an innovative air-breathing PEFC design. The piezoelectric microdiaphragm actuator served as an active air-flow channel within the fuel cell system. Outward actuator movement increased the cathode channel volume, effectively drawing air into the chamber. Inward movement of compressed air into the catalyst layer enhances the electrochemical reaction. Critically, PZT actuation produced more stable current output, stronger suction, and more effective water drainage. It also overcame concentration losses at optimal operating frequency, establishing the foundational viability of PZT-PEFCs.

Building on this foundation, Ma et al. [106] examined how rib design influences PZT-PEFC performance. A shallow rib was selected to reduce ohmic and concentration losses, as well as pressure drop. A transient

three-dimensional model was developed to compare ribbed versus non-ribbed configurations. The results confirmed that the ribbed cathode channel reduced ohmic losses and doubled current generation. Furthermore, at a higher PZT vibration frequency of 64 Hz, more oxygen was compressed into the catalyst layer. This enhancement of the electrochemical reaction resulted in a current density of 0.208 A/cm^2. Similarly, the role of the cathode "open area" was afterwards examined by Ma et al. [107] through a comprehensive three-dimensional theoretical model. Different cathode open-area ratios (80.5%, 63.2%, 47.9%, and 34.7%) were systematically evaluated. The cathode open-area ratio of 47.9% emerged as the optimal configuration among those tested. Lower vibration frequencies drew less air, leading to water vapour accumulation in the electrochemical reaction area. This accumulation eventually led to a decline in current output. Additionally, interdigitated anode flow-field designs outperformed serpentine and parallel configurations in terms of current density. Nozzle and diffuser geometries were subsequently incorporated to further refine airflow directionality and efficiency. In another study, Ma et al. [108] experimentally validated the PZT-PEFC—ND design, comprehensively confirming earlier theoretical predictions. The nozzle-diffuser configuration directed airflow into the cathode channel and prevented backflow without requiring valves. An optimal operating temperature of 323 K was identified to prevent membrane drying. The optimal vibration frequency was determined to be 180 Hz for sufficient air pumping. The PZT-PEFC—ND achieved a performance of 0.18 W/cm^2 without an external air-supply device.

Likewise, the aspect ratio and diffuser angle were further optimised through the pseudo-bipolar bi-cell PZT-PEFC—ND design. Ma et al. [109] demonstrated that an aspect ratio of 11.25 with a diffuser angle of 5° ensured smoother air intake. Lower aspect ratios resulted in insufficient actuation pressure, preventing effective water removal from the cathode. Larger aspect ratios induced a blocking phenomenon that restricted air intake into the cathode chamber. The bi-cell configuration achieved a performance 1.6 times that of the single cell. Ma et al. [110] subsequently confirmed these findings experimentally using the same bi-cell design parameters. Their experimental results corroborated that the bi-cell outperformed the single cell by the same factor. Performance in these studies was assessed through current-generation benchmarks against passive counterparts, reflecting this cluster's core metric. Ma et al. [111] further investigated the bi-cell using a reduced-nozzle-and-diffuser design. Operation at a stoichiometric ratio of 1.5 and a cell temperature of 50 °C effectively prevented concentration loss. Even with one degraded MEA, the bi-cell continued delivering non-negative power output. This resilience reinforced the viability of PZT-PEFC—ND bi-cells for future stack development. The maximum net power of the bi-cell was 0.7 W in that configuration. Ma et al. [112] complemented this by operating the bi-cell under various conditions with different intake modules. Parallel flow intake produced a maximum power of 0.283 W/cm^2, exceeding the series configuration's 0.263 W/cm^2. Asymmetric PZT amplitude was found to cause uneven airflow distribution across cathode chambers. This asymmetry resulted in differing current outputs between the two component cells. The PZT power consumption was identified as a critical factor when calculating net power output. Addressing this asymmetry challenge, Ma et al. [113]

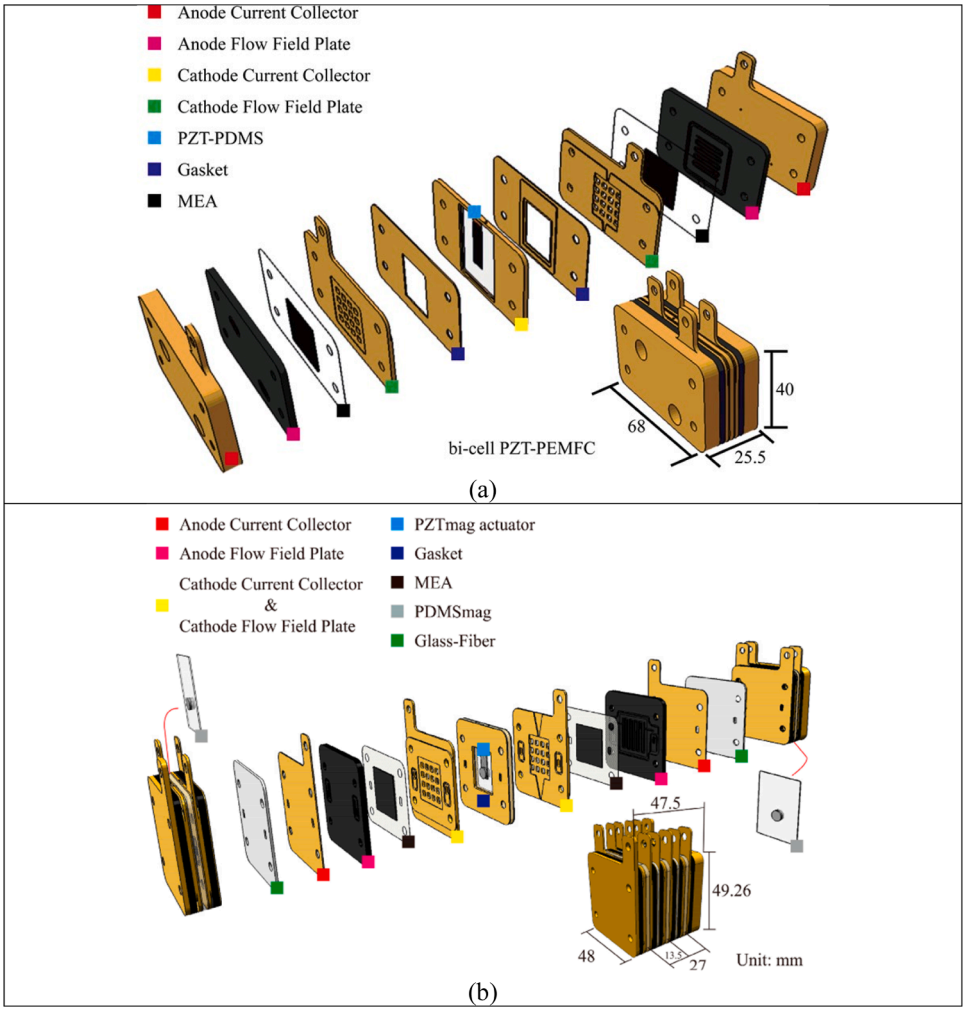


Fig. 9. (a) Structural schematic of the bi-cell PZT-PEFC system [113] and (b) layout of bi-cell PZTmag-PEFC stack configuration [114]. Published under open access.

developed a modified bi-cell with a groove-designed PZT actuator, as shown in Fig. 9(a). The actuator was enclosed in polydimethylsiloxane (PDMS) during curing to reduce uneven airflow between the two sides. This design achieved only a 0.7% difference in open-circuit voltage between the two cell sides. The new bi-cell module yielded a net power density of 0.1658 W/cm^2, exceeding previous iterations. Reducing volume and weight through gold-plated aluminium 6061 further enhanced its portability and practical appeal. Ma et al. [114] extended this design paradigm to a hybrid magnetically driven actuator integrated into a stack (Fig. 9(b)). Permanent magnets combined with a piezoelectric actuator simultaneously drove three air-breathing pumps using magnetic force. Six cathode flow field plates in a three-bi-cell stack were supplied with air simultaneously and uniformly. A nozzle-diffuser aspect ratio of 13.13 enabled uniform oxygen distribution to all membrane electrode assemblies. The stack achieved a maximum net power flux of 0.1925 W/cm^2 with 68% less volume. Weight reduction reached 76%, underscoring the compactness advantage of PZT-based actuation strategies.

Finally, it can be observed that Cluster 2 highlights the trend towards incorporating piezoelectric actuators in air-breathing PEFCs, without an external pump system, to regulate air flow and control the delivery of oxygen and water within the cathode assembly. Studies have shown that the combination of a PZT-based nozzle-diffuser, optimised rib structures, vibrating frequencies, and double-cell structures can significantly increase current production, power density, compactness, and robustness. Yet, the main concern for future research remains the difficulty of realising long-term, efficient, and uniform airflow actuation across multi-cells, while minimising actuator power and airflow imbalance. Also, although there is notable progress in improving the efficiency of oxygen transport, water management, and miniaturisation through PZT actuation strategies, the issues of longevity, energy dissipation, and scalability in incorporating piezoelectric actuators into multi-cell PEFC stacks still require significant consideration.

3.4.3. *Flow field design and numerical modelling of passive fuel cells*

Cluster 3 comprises themes including crosslinking, flow field, membrane electrode assembly (MEA), numerical model, open ratio, passive fuel cell, stack, and temperature distribution. This cluster focuses on research into the design, numerical analysis, and performance optimisation of passive fuel cells. A dual focus on structural and material design alongside computational modelling is evident throughout. The "flow-field" and "open ratio" terms are geometrically interlinked and govern the fundamental balance in passive fuel cell performance. Xing et al. [115] addressed this balance directly by optimising the geometry of air-breathing PEFCs using sequential quadratic programming. The simulations revealed that the optimum cathode open ratios were consistent across operating voltages. At 0.7 V and 0.4 V, optimal cathode open ratios of 49.8% and 49.5%, respectively, were identified. When the cathode-to-open ratio was fixed at 80%, the optimal anode-to-channel ratio was 34.7%. These findings highlight that systematic numerical optimisation can yield precise geometric configurations for passive fuel cells. Xu et al. [86] extended this investigation to air-breathing high-temperature PEFCs, experimentally and numerically examining multiple cathode flow-field configuration parameters. The study found that cathode end plate thickness and orientation had the least influence on performance. A slot-hole cathode flow field with a 75% open ratio achieved a maximum power density of 160 mW/cm^2 at 394 mA/cm^2. This result emphasises that open-ratio optimisation is configuration-dependent and must be evaluated within the broader flow-field geometry. Abdulrahman et al. [116] identified cathode GDL thickness as the most influential design parameter through Taguchi ranking. Notably, air-breathing PEFCs showed greater sensitivity to cathode GDL porosity than their conventional counterparts. This keen sensitivity was attributed to greater diffusion limitations and lower oxygen mass-transport coefficients in passive configurations. The MEA's material composition, particularly its membrane characteristics, is another central concern in this cluster. Chul Gil et al. [117] synthesised sulfonated polyethersulfone membranes with thermally crosslinkable endgroups for DMFCs. The study found that allyl-terminated variants achieved higher proton conductivity and methanol permeability than propargyl-terminated counterparts. The allyl-terminated membrane achieved a power performance of 23.5 mW/cm^2 in air-breathing semi-passive DMFC mode after activation. Crosslinking thus served both mechanical and electrochemical functions within the passive MEA context. Li et al. [118] further advanced crosslinked membrane design by using non-fluorinated polyamides with sulfonimide bifunctional bridges, as illustrated in Fig. 10. These bridges reinforced the membrane mechanically and facilitated proton-conduction pathways. The membranes with appropriate degrees of crosslinking exhibited water retention capacity exceeding that of commercial Nafion membranes. Maximum proton conductivity reached 0.139 S/cm at 80 °C, comparable to Nafion 117 performance.

Similarly, water management within the MEA remains a persistent challenge that numerical modelling can help address. In view of this, Xiong et al. [75] designed a dual-catalyst-layer cathode structure to enhance water management in air-breathing PEFCs. A hydrophilic inner Nafion layer-maintained proton transfer rates adjacent to the electrolyte membrane. A hydrophobic outer layer comprising Nafion and PTFE expelled excess water under high-current-density conditions. MPL thickness and GDL hydrophobicity were identified as significant parameters affecting air-breathing performance. This optimised MEA configuration improved water management across a range of current densities. Kumar et al. [119] approached MEA water management differently, proposing a dead-end hydrogen anode for intermediate water storage. A dynamically validated numerical water balance model predicted the effects of MEA configuration on resistance and performance. Their validated model showed that thin membranes combined with thicker GDLs enhanced membrane hydration and anode water-transfer rates. Varying current densities and cathode humidity levels were examined to systematically quantify the effects of membrane hydration. Temperature distribution and orientation effects on passive fuel cell performance were examined by [120]. Three different planar and ducted cathode designs with varying degrees of hydrophobisation were evaluated. Vertical channel orientation yielded higher performance across both planar and ducted cathode designs. Ambient temperature was found to significantly influence overall air-breathing fuel cell performance. Akyalçın [121] applied Taguchi experimental design to optimise MEA component combinations and the cathode current-collector geometry. The most influential parameters, in descending order, were cathode platinum loading, Nafion membrane thickness, cathode current collector opening geometry, and anode platinum loading. The optimal configuration achieved a power output of 98.5 mW/cm^2 at a cell voltage of 0.5 V. Pacheco et al. [122] proposed a novel millimetre-sized air-breathing PEFC with sequentially deposited layers, extending analysis to a two-cell stack. The design achieved a power density of 110 W/L and a specific power density of 2200 W/kg.

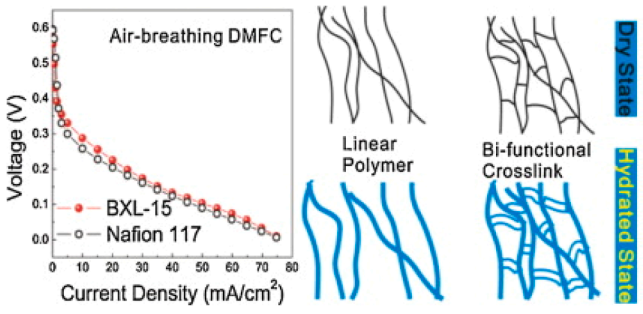


Fig. 10. Performance and structural comparison of BXL-15 and Nafion 117 membranes in an air-breathing direct methanol fuel cell [118]. Reprinted with permission from Elsevier.

Lin [123] further expanded the scope of numerical modelling to high-altitude UAV applications of air-breathing PEFCs. A modified Springer model with atmospheric corrections captured the effects of pressure, temperature, and humidity on membrane hydration. Simulations extended to 5000 m revealed reductions in power and efficiency of approximately 70% and 15%, respectively. Active humidification and optimised flow channels were found to recover >50% of altitude-induced losses.

In conclusion, Cluster 3 highlights the optimisation of passive air-breathing PEFCs through innovations in flow-field design, membrane electrode assembly materials, water management, and the modelling of heat and transport processes. The results from the reviewed studies showed that optimised cathode open-area ratios, cross-linked membrane materials, hydrophobic-hydrophilic membrane-electrode assemblies, and computational stack design approaches can lead to significant improvements in proton transport and passive fuel cell efficiency. Nevertheless, the main engineering obstacle remains the inability to simultaneously achieve uniform temperature distribution, water management, and oxygen transport in passive fuel cells across various operating environments.

3.4.4. Performance characterisation and thermal-water diagnostics of portable air-breathing fuel cells

Cluster 4 comprises the following themes: air-breathing fuel cell, cell characteristics, hydrogen, performance, portable fuel cell, thermal imaging, and water transport. This cluster is primarily focused on experimental characterisation and performance evaluation of air-breathing fuel cells, with a strong emphasis on portable applications. These themes imply the need for research quantifying how air-breathing fuel cells behave under real operating conditions. Observable physical phenomena, especially thermal and water transport, are linked to electrochemical performance throughout this body of work. The keywords "air-breathing" and "portable fuel cell" establish the application context: compact, lightweight hydrogen power sources that draw oxygen directly from ambient air. In this context, Obeisun et al. [87] demonstrated this by developing open-cathode polymer electrolyte fuel cells with printed-circuit-board flow-field plates. Thermal imaging revealed lower temperatures in the middle of the fuel cell, indicating plate bowing. This bowing reduced local current density non-uniformly across the active area. A trade-off between competing design factors yielded a maximum area-specific power density of 250 mW/cm^2. This study demonstrated that thermal imaging is far more informative than point thermocouple measurements for diagnosing operational non-uniformities. Obeisun et al. [124] further exploited thermal characterisation by investigating the dynamics of water droplets on GDL surfaces. The wettability analysis revealed that droplets maintained a fixed contact diameter until transitioning from a non-wetting to a wetting regime. After this transition, the contact diameter reduced rapidly until complete evaporation occurred. GDL thermal characterisation attributed temperature variation to changes in emissivity and uneven surface thermal gradients. Droplets introduced at room temperature cooled the surrounding GDL material. This cooling effect was more significant at higher GDL operating temperatures. These insights are relevant to passive open-cathode cells, in which no active water-removal mechanisms exist.

Similarly, “water transport” in this cluster suggests a dual challenge of membrane hydration and flooding prevention in air-breathing configurations. For instance, Hamel and Fréchette [125] reported that anode humidification control is a determining factor in the performance of miniature air-breathing PEFCs. An analytical one-dimensional model revealed simultaneous anode drying and cathode flooding at low temperatures. This simultaneous occurrence was attributed to dominant electro-osmotic drag under those conditions. Experimentally, injecting water into the anode at three times the cathode production rate significantly improved high-current-density performance. The current and maximum power densities were increased by 100% and 30%, respectively, using this method. Chaparro et al. [126] employed thermal neutron radiography to visualise the distribution of water in a passive PEFC for portable hydrogen systems. The cell operated under passive conditions with an air-breathing cathode and a dead-end dry hydrogen anode. A columnar cathodic plate was found to favour the mobility and elimination of water droplets from the cathode surface. In the vertical orientation, natural convection and gravity combined to effectively reduce the cathode water content. The vertically oriented cell provided steady power densities above 100 mW/cm^2 over extended periods of operation. In the horizontal orientation, reduced gravity reduced peak power density by 17%. Anode flooding caused cell failure after only 5 mg/cm^2 of water production when the cathode faced upward. Folgado et al. [17] extended this columnar plate concept, demonstrating that it doubled the mass-transport-limited current compared to a planar perforated plate. The columnar plate reduced water saturation in the cathodic GDL, significantly improving overall gas transport. It also reduced the dependence of cell performance on orientation, enhancing applicability across diverse portable uses. Peak power density of the passive air-breathing cell with the columnar plate was only 15% below that of a conventional cell. This modest reduction anticipates higher specific power density for the complete portable system.

Likewise, cathode design geometry and its effect on cell characteristics were systematically examined by Kumar and Kolar [127]. In this study, ducted and ribbed cathode designs were compared using current density, species distribution, temperature, and water transport coefficient metrics. The ribbed cathode achieved a peak power density of 232 mW/cm^2, exceeding that of the ducted cathode (205 mW/cm^2). The ribbed cathode's current density reached 690 mA/cm^2, outperforming the ducted design's 430 mA/cm^2. Vertical cell orientation consistently yielded the best performance across both cathode designs. Kumar and Kolar [128] further comprehensively examined the effect of cathode channel dimensions on air-breathing PEFC performance. Performance improved with increasing channel width and depth and with decreasing cell height. A maximum power density of 240 mW/cm^2 was achieved with a 4 mm channel width and 6 mm channel depth. At channel depths below 2 mm, buoyancy-induced flow resistance severely limited performance. Above a depth of 2 mm, GDL diffusion resistance became the dominant limiting factor at low current densities. Flooding was identified as a risk at low current densities, while membrane dehydration dominated losses at high current densities. Chen et al. [129] addressed gas diffusion and water transport limitations through a novel graphene-on-carbon-cloth microporous layer. Lotus leaf graphene grown on carbon fibre substrates formed a highly porous, electrically conductive microporous cathode layer. The resulting structure demonstrated high structural stability, minimal mass transfer resistance, and cost-effectiveness. The PEFC with this configuration achieved a maximum current density of 1480 mA/cm^2 and a peak power density of 398.9 mW/cm^2. After 12 h of high-current-density operation, current density declined by only 7.94%, confirming excellent durability.

This cluster highlights the experimental characterisation and operational evaluation of portable air-breathing PEFCs, with a primary focus on thermal performance, water transport phenomena, cathode geometry, and orientation dependence. All these findings suggest that optimal cathode design and the use of microporous materials can effectively enhance the performance of portable fuel cells by improving water removal, gas diffusion, and current distribution. However, the major unsolved issue lies in passively maintaining stable, uniform water and thermal management.

3.4.5. Durability, high-temperature operation, and free convection across fuel cell types

Cluster 5 comprises terms including durability, flooding, free convection, high temperature, microbial fuel cell, and polybenzimidazole. This cluster brings together research themes related to long-term durability and operational robustness across multiple fuel cell types. The use of “polybenzimidazole” indicates a focus on higher-temperature proton-exchange membranes that offer significant durability advantages. Wu

et al. [130] demonstrated the performance potential of high-temperature air-breathing PEFCs using phosphoric acid-doped polybenzimidazole membranes. The design achieved a peak power density of 220.5 mW/cm^2 at 200 °C without any water-management systems (Fig. 11). This result was comparable to that of conventional Nafion membranes operating at much lower temperatures. The oxygen-transfer limitation at this higher temperature resulted in a limiting current density of approximately 700 mA/cm^2. A short-term durability test at 200 mA/cm^2 and 180 °C showed a gradual decrease in voltage accompanied by an increase in internal resistance. The continuous-operation degradation rate of 58.32 μV/h was substantially lower than that observed during start-stop cycling. This finding underscores the durability advantage of sustained operation over intermittent cycling in high-temperature configurations.

Flooding, though less prevalent at high temperatures, remains relevant during cold-start and transient operation phases. Paquin and Fréchette [21] addressed this through a comprehensive analytical and experimental investigation of flooding and dry-out phenomena in air-breathing PEFCs. The authors found that the ratio of thermal resistance to mass transport resistance determines which limiting phenomenon manifests. Decreasing this ratio beyond a critical threshold induced flooding, while excessively increasing it caused membrane dry-out. Too high a thermal resistance or too low a mass-transport resistance was found to reduce the limiting current density by accelerating dry-out. This work established a systematic framework for balancing thermal rejection and mass transport to optimise the performance of free-convection fuel cells. Babcock et al. [131] further investigated the cathode optimisation challenge for passive air-breathing fuel cells relying on free convection. The study examined the trade-off between mass-transport and ohmic electrical-resistance losses. High-frequency resistance measurements and fast-scan polarisation measurements were used to interrogate both loss mechanisms independently. Pressure-sensitive foils established a quantitative link between pressure distribution and cell resistance across configurations. Deploying a sufficiently stiff yet highly porous contact layer significantly reduced contact resistance while incurring minimal additional mass-transport losses. This strategy enabled cathode structures with open areas exceeding 90%, thereby enhancing performance, particularly at high current densities.

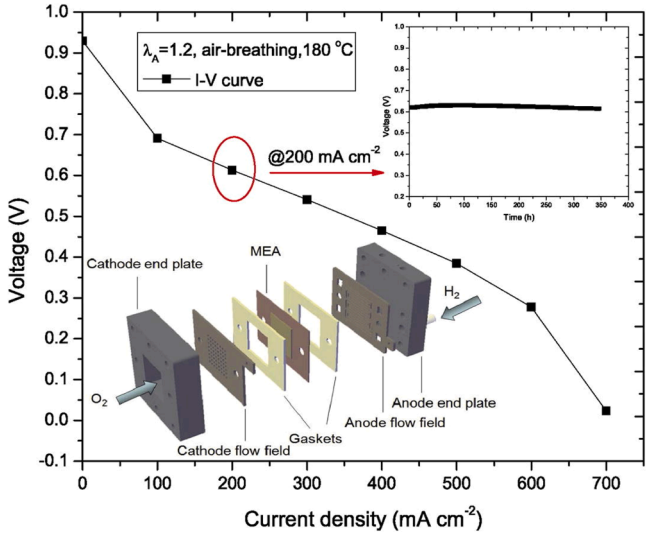


Fig. 11. Polarisation (I-V) curve of an air-breathing PEM fuel cell operating at $\lambda_a = 1.2$ and 180 °C, highlighting performance at 200 mA/cm^2; inset shows voltage stability over time at 200 mA/cm^2, and the lower schematic illustrates the exploded configuration of the single-cell assembly, including cathode/anode end plates, flow fields, MEA, and gaskets. Reprinted from [130] with permission from Elsevier.

Similarly, polybenzimidazole membranes are also notably featured in microbial fuel cell research within this cluster. Bahlakeh et al. [132] fabricated and characterised phosphoric acid-doped polybenzimidazole (PBI) membranes for air-breathing microbial fuel cell applications. Molecular dynamics simulations provided an in-depth understanding of membrane structure-property relationships, revealing strong phosphoric acid-water interactions. Acid-doped PBI membranes showed reduced oxygen permeability of 0.36 barrers, compared to 1.78 barrers for Nafion. The PBI-based MFC produced a higher open-circuit voltage of 471 mV and a maximum power density of 74.2 mW/cm^2. In contrast, Nafion membranes achieved only 396 mV and 48.6 mW/cm^2 under equivalent conditions. The higher electrochemical performance was attributed to considerably lower ion and oxygen diffusion coefficients predicted by molecular dynamics simulation. Hasani-Sadrabadi et al. [133] pursued an alternative approach to improving the performance of air-breathing MFCs through nanocomposite proton-exchange membranes. Sulfonated poly(ether ketone) and montmorillonite nanocomposite membranes were systematically developed and evaluated. Although montmorillonite (MMT) incorporation reduced both oxygen permeability and proton conductivity individually, overall membrane selectivity improved. The optimised sulfonated poly(ether ketone)-70/MMT-3 wt% nanocomposite membrane delivered approximately 40% more power than Nafion 117. Hou et al. [134] evaluated the durability of potassium hydroxide (KOH)-doped polybenzimidazole membranes in an air-breathing alkaline direct ethanol fuel cell. Ex situ durability testing showed ionic conductivity degrading from 0.023 S/cm to 0.01 S/cm over 100 h of operation. The degradation rate was determined to be 1.3×10^{-4} S/cm h under these conditions. In situ testing demonstrated stable operation above 0.3 V for 336 h, confirming acceptable long-term durability. Density functional theory was additionally employed to explore interactions between KOH and the PBI matrix. Schönberger et al. [135] investigated partially fluorinated poly(arylene ether)s as ideal membrane materials for fuel cell applications. Sulfonation routes and degradation behaviour were characterised by gel permeation chromatography across the polymer series. The most promising polybenzimidazole blend membrane was evaluated in an air-breathing DMFC and yielded improved performance compared to Nafion 105. Kazemi et al. [136] demonstrated the viability of passive air-breathing flat-plate microbial fuel cells as economically competitive configurations. Continuous operation with a proton exchange membrane achieved a peak power density of 18 W/m^3. However, greater operational stability during continuous mode resulted in a 2.5-fold increase in peak power density to 44 W/m^3. It can be observed that research carried out in Cluster 5 focused on the long-term durability, thermal robustness, and operational stability of air-breathing fuel cells using polybenzimidazole-based membranes, as well as passive free-convection fuel cell systems. The research shows that advanced membrane chemistry, optimisation of thermal-mass transport balance, and highly porous cathode structures can increase resistance to flooding, reduce oxygen crossover, and provide better electrochemical stability for PEFCs and microbial fuel cells. Despite advancements, a key engineering challenge remains: ensuring long-term membrane conductivity and structural integrity under combined thermal, chemical, and cyclic stresses. This challenge is critical to maintaining efficient water and mass transport in passive air-breathing configurations.

3.4.6. Miniaturisation, flow channel design, and rapid prototyping for direct methanol fuel cells

Cluster 6 contains terms such as DMFC, flow channel design, fuel buffer/distributor, fuel cell performance, miniature fuel cell, and rapid prototyping. This cluster focuses on the design and fabrication of miniaturised DMFCs, with an emphasis on flow-channel geometry and modern manufacturing approaches. Direct methanol is a promising liquid hydrogen carrier due to its ease of storage and transport for mobile and marine applications [137]. Its efficient, cost-effective conversion to H_2 is essential for fuel-cell applications as hydrogen demand rises

toward net-zero emissions[138–140]. Methanol's high energy density and ambient-condition storage make DMFCs suitable for portable electronics applications [141]. The term "rapid prototyping" in this cluster implies that researchers focus on product development to create a physical or digital model (a prototype) of a design idea, which can be tested, evaluated, and improved before full-scale production. For instance, Chen et al. [142] developed a 10-cell air-breathing miniature PEFC stack using rapid prototyping technology, with acrylonitrile-butadiene-styrene as the primary flow-field plate material. This approach enabled fast commercial design, low-cost manufacturing, and compatibility with plastic injection moulding at scale. The compact stack measured only 6 cm × 6 cm × 0.9 cm, demonstrating the dimensional advantages of rapid prototyping for miniaturised systems. Under free convection, the parallel- and series-connected configurations achieved peak power densities of 99 mW/cm^2 and 92 mW/cm^2, respectively. Forced convection through an electric fan improved these values to 123 mW/cm^2 and 105 mW/cm^2, respectively. Forced convection additionally enhanced the uniformity of the output voltage distribution across the stack. Chen et al. [143] extended this rapid prototyping approach to more geometrically complex fuel cell designs. Honeycomb methanol reservoirs, complex cathode structures, and monopolar air-breathing PEFC flow distributors were fabricated using three-dimensional CATIA software and rapid prototyping. These complex geometries were considered nearly impossible to manufacture through traditional machining methods. The feasibility of a complex pseudo-bipolar DMFC stack design was also verified experimentally in this study. Both DMFC and PEFC designs benefited from the manufacturing flexibility that rapid prototyping uniquely enabled.

The "flow channel design" in this cluster is a central engineering challenge within miniaturised DMFC and PEFC configurations. To address this, Zhang et al. [144] developed a three-dimensional model for an air-breathing micro-DMFC with an extended anode catalyst region on channel walls, as displayed in Fig. 12. It was observed that fuel cells with anodes on both the membrane and channel walls did not outperform membrane-only anode configurations. The overall power generation was found to be primarily limited by methanol electrode reaction kinetics rather than methanol transfer rates. Increasing the anode catalyst loading on the channel walls decreased the cathode potential by increasing ohmic losses. Reducing channel dimensions simultaneously decreased ionic resistance and increased methanol concentration at the anode. Rao et al. [145] found that the cross-sectional geometry of the flow channel influences the performance of passive micro-DMFCs. For example, the trapezoidal cross-section flow channels increased power density to 6.64 mW/cm^2, nearly doubling the 3.9 mW/cm^2 achieved with rectangular channels at a methanol concentration of 7 M. This enhancement was attributed to improved fuel distribution characteristics of the trapezoidal geometry. An optimal methanol concentration of 7 M was identified for maximising both open-circuit voltage and power density. These geometric insights were suggested to be broadly applicable beyond DMFCs to other PEM-based fuel cell types.

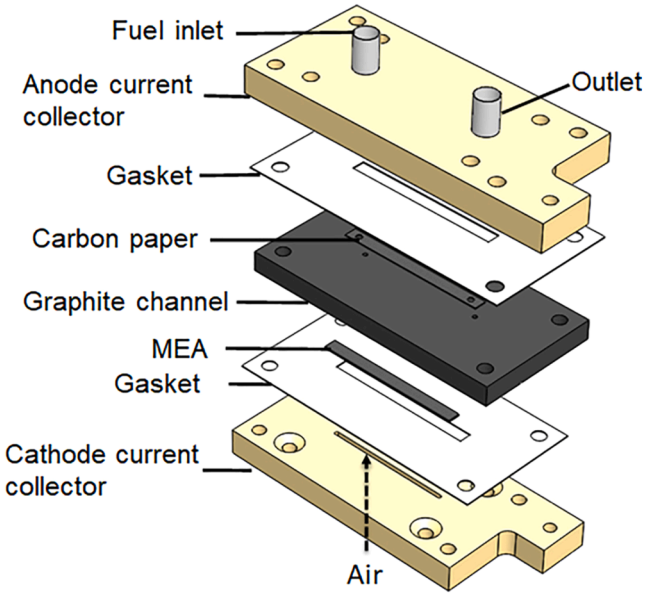


Fig. 12. Experimental air-breathing μDMFC configuration [144]. Published under open access.

Thermal management and stack-level flow channel configuration are significant performance determinants in this cluster. For instance, Yan et al. [91] investigated thermal management of high-powered air-breathing PEFC stacks through varying cathode flow channel configurations. A combination of 50% and 58.3% opening ratios within the stack simultaneously reduced temperature and enhanced uniformity of temperature distribution. More stable, improved stack performance resulted directly from this optimised opening-ratio combination. Membrane material design contributes to the performance of miniature fuel cells in this cluster. Li et al. [146] synthesised 3D-branched rigid-flexible hybrid sulfonated polyamide membranes for DMFC applications. The 3D rigid-flexible hybrid structure demonstrated 20% higher conductivity than the wholly rigid version at a comparable ion-exchange capacity. The best-performing membrane achieved a conductivity of 0.162 S/cm, exceeding Nafion 117 at 80 °C. An air-breathing DMFC assembled with this membrane exhibited 12% higher power density than Nafion-based devices with 4 M methanol. Alipour et al. [147] evaluated potassium-doped Nafion 117 membranes in passive direct methanol alkaline fuel cells. The alkaline configuration achieved an open-circuit voltage approaching 1 V, substantially exceeding the 0.45 V of passive DMFC under acidic conditions. Potassium-doped membranes exhibited lower methanol crossover permeability and higher mechanical and thermal stability. UAV and high-altitude applications further extend the context for evaluating miniature fuel cell performance. González-Espasandín et al. [69] compared air-breathing hydrogen PEFCs and passive DMFCs under atmospheric UAV flight conditions. Low pressure affected PEFC/hydrogen performance more severely than DMFC performance across the altitudes examined. The atmospheric relative humidity influenced PEFC performance at high cell temperatures, whereas DMFC performance was barely affected. Operation at low-to-medium altitudes remained feasible for both fuel cell types in UAV propulsion systems. Lin [123] further confirmed altitude-dependent performance reductions for air-breathing PEFCs, with power and efficiency dropping approximately 70% and 15%, respectively, at 5000 m. Active humidification and optimised flow channels were shown to recover >50% of altitude-induced performance losses. Finally, it was observed that Cluster 6 addresses the miniaturisation and manufacturability of air-breathing DMFC and PEFC systems by adopting advanced flow-channel designs, rapid prototyping methods, and innovative membrane structures for portable and UAV applications. Overall, the studies showed that optimised flow channels, compact stack arrangements, additive manufacturing methods, and membranes with enhanced characteristics can be used to improve the performance of miniature fuel cells. Despite advancements, a key engineering challenge persists: achieving high power density, ensuring uniform reactant distribution, maintaining thermal stability, and minimising methanol crossover in compact stack architectures. This challenge must be addressed while also ensuring scalability and reliable performance across varying environmental conditions and altitudes.

3.4.7. Gas diffusion layer and microporous layer design for liquid water management in PEFCs

Cluster 7 contains themes including gas diffusion layer, geometry variation, liquid water removal, microporous layer, modelling, and water management. This cluster addresses one of the most fundamental and technically challenging aspects of PEFC design: the structure and

function of the GDL and its role in water management. The GDL distributes reactant gases, conducts electrons, and manages the transport of liquid water produced at the cathode. To assess the performance of GDL, Obeisun et al. [124] investigated water droplet dynamics on GDL surfaces using wettability and thermal characterisation. Droplets maintained a fixed contact diameter until transitioning from a non-wetting to a wetting regime. After this transition, the contact diameter reduced rapidly until complete evaporation occurred. GDL thermal characterisation attributed temperature variation to changes in emissivity and uneven surface thermal gradients. Droplets introduced at room temperature cooled the surrounding GDL material, with a greater effect at higher operating temperatures. These insights are relevant to passive open-cathode cells, where water removal occurs solely by evaporation. Xiong et al. [75] addressed water management through a dual-catalyst-layer cathode structure combined with optimised GDL and microporous layer (MPL) parameters. A thin, hydrophilic Nafion inner layer maintained high proton-transfer rates adjacent to the electrolyte membrane. A hydrophobic outer layer comprising Nafion and PTFE expelled excess water under high-current-density conditions. MPL thickness and GDL hydrophobicity were identified as influential parameters on air-breathing performance. This dual-layer MEA configuration improved water management under varying current densities. Choi et al. [148] focused specifically on optimising the PTFE content in the microporous layer of passive-type air-breathing PEFCs. Physical MPL properties were evaluated using field-emission scanning electron microscopy and contact angle measurements. Both 5% and 20% PTFE content yielded higher performance than the other configurations tested. The optimal PTFE content was determined through combined performance and durability evaluations under passive operating conditions. These findings were expected to enhance the suitability of fuel cells as a compact, portable power source.

In other studies, Karst et al. [149] investigated the role of cathodic cover opening ratio on water management in micro air-breathing PEFCs. By varying the cover configuration, internal resistance ranged from 0.33 Ω cm^2 to 4.0 Ω cm^2. A 5% opening ratio cover allowed cells at 10% relative humidity to reach current densities comparable to those at 70% relative humidity without a cover. Complete cover closure allowed accumulated water to remain in place, enabling current densities of 800 mA/cm^2 after eight minutes. The maximum back-diffused water, 33% of the total water product, was observed at the 5% opening ratio configuration. In another study, Karst et al. [150] reported that cathodic crack structures simultaneously reduced both cell drying and flooding. Cracks in the cathodic catalytic layer allowed the gold current collector to contact the electrolyte, maintaining membrane hydration. Current density increased from 28 mA/cm^2 to 188 mA/cm^2, a 570% improvement, under low-humidity conditions with a cracked cathode network. Under flooding conditions at 70% relative humidity, cracked cathode cells showed higher current densities than non-cracked configurations.

It is worth noting that novel GDL materials and geometries are a significant sub-theme within this cluster. Wei et al. [74] developed an ultra-thin, flexible carbon nanotube GDL using a simple three-step preparation process at only 350 °C. The resulting GDL was <40 μm thick, approximately one-sixth the thickness of commercial GDLs. The air-breathing PEFC equipped with this GDL achieved a volume-specific power density of 15,600 W/L and a weight-specific power density of 9660 W/kg. Stack volume-specific power density was estimated to improve by >60% by replacing conventional GDLs with this material. Dang et al. [151] engineered natural wood-derived macroporous carbon sheets with three-dimensional perpendicular channels as GDLs (Fig. 13). These wood-derived GDLs achieved a current density of 0.139 A/cm^2 at 0.6 V and a maximum power density of 0.102 W/cm^2. Performance was comparable to air-breathing PEFCs with higher platinum loadings. Chai et al. [29] designed a biomimetic, integrated GDL inspired by alveolar structures to improve the operation of air-breathing PEFCs. The biomimetic GDL delivered a peak power density of 148.5 mW/cm^2, outperforming the conventional GDL's 109.8 mW/cm^2. At a cathode wind speed of 4 m/s, the biomimetic GDL cell declined by only 2.38% after 100 hrs of operation. In contrast, the traditional fuel cell declined by 64.5% within just 10 min under identical conditions. The fuel cell stack with this GDL exhibited 17.5 W at 3.17 V with only a 11.8% decline after 200 h. However, computational modelling plays a major role in understanding GDL transport mechanisms. Fu et al. [152] developed pore-scale Lattice Boltzmann models for the air-breathing cathode of membraneless microfluidic fuel cells. GDL microstructure was found to play an important role in oxygen transport and cathode reaction. Oxygen penetrated the electrolyte microchannel, forming a thin dissolved-oxygen layer over the catalyst layer. Fabian et al. [79] reported a passive water-management design using a hydrophilic,

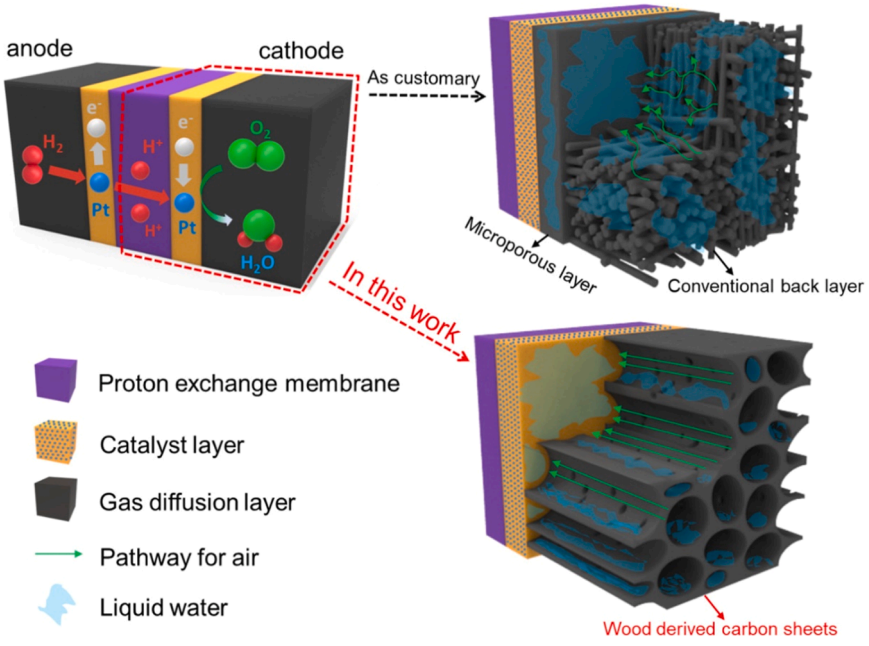


Fig. 13. Natural wood-derived macroporous carbon sheets as a gas diffusion layer in a proton exchange membrane fuel cell, illustrating enhanced air transport and water management. Reprinted from [151] with permission from Elsevier.

electrically conductive wick. This design operated under severe flooding conditions at 10 °C and 80% relative humidity for up to 6 hrs without performance loss. Fabian et al. [153] extended this concept by using an electroosmotic pump to actively remove water from the wick. The electroosmotic pump consumed <2% of fuel cell power and eliminated cathode flooding. Folgado et al. [17] reported that a columnar cathodic plate doubled the mass-transport-limited current compared to a planar perforated plate by decreasing the GDL water saturation. Zeidan et al. [154] developed a steady-state macroscopic model that captures the sensitivity of water content to temperature and relative humidity in micro air-breathing fuel cells. Chen et al. [129] achieved a peak power density of 398.9 mW/cm^2 using lotus leaf graphene grown on a carbon fibre cathode microporous layer.

This cluster addresses the optimisation of gas diffusion layers and microporous layers for water and oxygen management and the durability of air-breathing PEFCs, utilising experiments, innovative materials, and multiscale modelling tools. Overall, research indicates that specific geometry configurations of GDL, hydrophobic-hydrophilic surface treatment, bio-inspired designs, electrical conductive wicking structures, and improved carbon-based materials have potential in enhancing liquid water drainage, membrane hydration, and operational durability. Nonetheless, the main problem that remains to be solved from an engineering perspective is the design of GDLs and MPLs that can simultaneously achieve efficient oxygen diffusion, rapid removal of liquid water, and sufficient thermal and mechanical strength without increasing the risk of flooding. Additionally, despite the evident advantages in water handling and power density offered by innovative designs like advanced GDL/MPL systems and new materials, including carbon nanotube composite, biomimetic, graphene-coated, and wood-derived structures, their stability, producibility, and reliability at the stack-level operation have not been sufficiently validated.

3.4.8. Multiphase modelling, air contaminants, and wettability in ab-PEFCs

Cluster 8 comprises a 3D multiphase model, air contaminants, contact angle, open area ratio, and ab-PEFC. This cluster focuses on computational modelling of air-breathing PEFCs, with specific attention to ambient air quality, surface wettability, and geometric parameters. The 3D multiphase model indicates the methodological basis of this cluster's computational research programme. Several authors have applied the multiphase model in the research domain of air-breathing PEFCs. For example, Liu et al. [16] developed a 3D multiphase model of single air-breathing PEFCs that incorporates contact resistance between the GDL and the bipolar plate. Anisotropic thermal conduction and electric conductivity in through-plane and in-plane directions were both accounted for in this model. The model achieved good grid independence and strong agreement with experimental polarisation curve data. The single PEFC with an optimal open-area ratio of 55% achieved a peak power density of 179.3 mW/cm^2. For a ten-cell fuel cell stack, the anode-window flow field configuration achieved higher peak power density than the anode-serpentine flow field. This model was presented as a practical tool to assist in the development of high-performance air-breathing PEFC stacks. Sun et al. [155] extended 3D multiphase modelling to investigate the influence of GDL compression rate on air-breathing PEFC performance. Anisotropic electric and thermal conductivities in both in-plane and through-plane GDL directions were incorporated alongside contact resistance between GDL and bipolar plate. The study found that the optimal open-area ratio for maximum performance varies with GDL compression rate. Generally, higher compression rates corresponded to larger optimal open-area ratios across the configurations examined. This interaction between GDL compression and open-area ratio highlights the geometric complexity governing ab-PEFC performance. Ma et al. [107] determined that a cathode open-area ratio of 47.9% outperformed those of 80.5%, 63.2%, and 34.7% in PZT-PEFC configurations. Lower vibration frequencies were found to draw less air into the cathode channel under those conditions. This insufficient airflow led to the accumulation of water vapour in the electrochemical reaction area over time. The resulting water accumulation ultimately led to a progressive drop in current output, underscoring the interaction between the open area ratio and water vapour management.

Similarly, “contact angle” and “surface wettability” are other significant design variables within this cluster. Ous and Arcoumanis [156] investigated water droplet formation in air-breathing PEFCs from two orthogonal viewpoints. The observations revealed that receding and advancing droplet contact angles during formation were almost identical. Both contact angle values increased in a trend similar to droplet height as droplet diameter increased. These findings provide fundamental insights into the wettability dynamics that govern liquid water behaviour within cathode channels. Baroutaji et al. [78] explored open-pore cellular foam as an alternative cathode current collector for air-breathing PEFCs. PTFE coating was applied to the foam to enhance its hydrophobicity and corrosion protection in the fuel cell environment. Contact angle and electrochemical polarisation tests confirmed that PTFE-coated foam outperformed uncoated foam in terms of cell performance. Open-pore cellular foam was subsequently used to build an air-breathing PEFC stack, demonstrating its scalability.

Likewise, “air contaminants” are an underappreciated but highly significant challenge for ab-PEFCs exposed to ambient air. Biesdorf et al. [157] conducted a unique long-term outdoor study of air-breathing fuel cells across three sites in Germany. Sites ranged from a highly polluted highway location to a virtually pure air site at 1205 m altitude. Fuel cells operated continuously for over 4500 h across all sites under varying weather and pollution conditions. The impact of real air contaminant mixtures on cell voltage degradation was found to be mostly reversible. However, unregulated cell temperature prevented the isolation of irreversible contaminant-induced degradation from dominant temperature effects. Faber et al. [158] systematically assessed the effects of specific airborne particulate contaminants and gaseous impurities on ab-PEFC performance. Contaminants examined included oxalic acid, soot, ammonium sulfate, sodium chloride, and nitrogen oxides. NOx at 10 ppmv degraded fuel cell performance by 22% within four hours of exposure. A 70% partial recovery was subsequently observed within an additional 11 h after NOx removal. Oxalic acid and ammonium sulfate showed measurable performance degradation, while sodium chloride and soot showed negligible effects. Because diffusion governs aerosol particle transport into passive ab-PEFCs, only a minor fraction of total particle mass is actually deposited inside the cell. Mitra et al. [159] specifically addressed CO contamination, proposing a combined range and proximate sensing approach for smoke plume tracking near-field-deployed air-breathing PEFCs. Platinum catalyst poisoning by CO was identified as a source of irreversible electrode damage. The simulation results quantified the dynamic changes in PEFC electrode resistance as a function of CO concentration. Drops in fuel cell output power as CO2 mass fraction and ambient humidity increased were also quantified. The proposed threat prediction approach aimed to enhance the lifetime of field-deployed air-breathing PEFCs by forecasting and mitigating contaminant exposure. Finally, it was observed that this cluster focuses on advanced computational modelling and the environmental interaction effects in air-breathing PEFCs. It highlights key areas such as multiphase transport behaviour, surface wettability, geometric optimisation, and the impact of ambient air contaminants. The studies showed that 3D multiphase models, optimised open-area ratios, and tailored surface hydrophobicity can enhance the understanding of coupled thermal, electrical, and water-transport phenomena in ab-PEFCs under realistic conditions. Despite these advancements, a major engineering challenge persists: maintaining stable long-term performance in air-breathing PEFC systems under dynamically changing environmental conditions, especially when contaminant exposure, surface wetting behaviour, and multiphase transport interact simultaneously within passive architectures.

3.4.9. Flexible and material architectures for next-generation fuel cell current collectors

Cluster 9 contains terms including CNT membrane, copper, current collector, flexible, and fuel cell. This cluster highlights a cutting-edge area of research focused on current-collector design, advanced materials, and flexible fuel-cell configurations. The cluster intersects advanced manufacturing, nanotechnology, and electrochemical engineering. The current collector is a fundamental component that conducts electrons generated at the anode or cathode to the external circuit. Conventional current collectors are typically rigid, heavy, and relatively expensive, machined graphite or stamped-metal bipolar plates. Research within this cluster thoroughly challenges these conventions through alternative materials and novel architectural configurations. The term "flexibility" in this context could perhaps be the most impactful research direction within this cluster's scope. In this context, Chang et al. [160] fabricated an air-breathing, flexible fuel cell using PDMS endplates for both the anode and the cathode. Metallic films were sputtered onto patterned PDMS substrates to create integrated current collectors within the flexible structure. The power density of the bent cell at a curvature of approximately 1.8 m^{-1} decreased by about 30% compared to the non-bent configuration. This feasibility study demonstrated that PDMS-based flexible fuel cells can maintain measurable electrochemical performance under mechanical deformation. Park et al. [161] further advanced the design of flexible fuel cells with a rollable, ultra-light, ultra-thin PEFC configuration. A thin flow-field plate was fabricated via thermal imprinting and integrated with a laser-machined metal current collector. The resulting air-breathing fuel cell measured only 0.992 mm in thickness and weighed just 2.23 g. It demonstrated a total power of 508 mW with <10% performance degradation after 200 repeated severe bending cycles. This high power-per-weight ratio of 0.228 W/g, alongside robust bending durability, has not been reported in the literature.

Similarly, the term "carbon nanotube (CNT) membranes" is among the most significant material innovations examined in this cluster. Ning et al. [162] found that the current-collector configuration affects the performance of flexible air-breathing PEFCs with large active areas. Performance increased >8-fold by adjusting the current collector direction alone, without any other modifications. These performance differences were attributed to variations in the lengths of electron-transfer pathways across different collection configurations. Flexible PEFCs with thicker carbon nanotube membrane current collectors, which have lower electrical resistance, demonstrated superior power delivery capability. This finding emphasised that CNT membrane conductivity is a critical determinant of large-area flexible fuel cell performance. Copper-based current collectors and their corrosion management are also significant themes within this cluster. Kuan et al. [163] proposed a novel lightweight planar current collector for low-power, portable PEFC applications. The design integrated a substrate, an electrical conduction layer, and a corrosion-resistant layer into a single, compact assembly. The FR-4 served as the substrate material, with a thin copper film deposited by thermal evaporation as the electrical conduction layer. A graphene thin film applied by spin coating and vacuum oven processing provided the corrosion-resistant layer. The assembled current collector was successfully tested in both forced-convection and self-air-breathing cathode configurations. In the work of Lee et al. [164], the authors developed multilayer metal-coated polycarbonate monopolar plates as an alternative current collector approach for portable fuel cells. The multilayer structure comprised electroless-plated copper, electroplated copper, nickel, and a gold layer to ensure adhesion, conductivity, corrosion resistance, and cost-effectiveness. Through selective plating on polycarbonate plates, air-breathing stacks were assembled without requiring additional, separate current-collector components. The planar stack achieved an open-circuit voltage of 2.85 V and a maximum power of 7.18 W. Suseendiran et al. [165] investigated cylindrical PEM fuel cell architectures with semi-cylindrical cathode current collectors as an alternative to conventional planar designs. Cylindrical configurations eliminated the need for bipolar plates and precisely machined flow channels simultaneously. Gold coating of current collectors reduced contact resistance from 75 mOhm to 30 mOhm progressively through design refinements. Optimising cathode rib width and hydrogen flow rate yielded a cumulative 40% increase in peak power output. The optimised cell delivered 400 mA/cm^2 at 0.6 V and a peak power of 2 W under air-breathing conditions.

Furthermore, the catalyst layer composition and its influence on current generation are also examined within this cluster. For example, Xiong et al. [166] adjusted the pore structure and hydrophilicity of air-breathing cathodes using dual catalysts of Pt/C and Pt/CNT. A Pt/C-to-Pt/CNT ratio of 54:46 in the cathode catalyst layer achieved a current density of 290 mA/cm^2 at 0.6 V, representing a 14% improvement over MEAs without Pt/CNT inclusion. Carbon nanotubes enhance charge transfer, electrical conductivity, and gas permeability due to their tubular, porous structure. Mashkani et al. [167] developed an Fe-ZIF-polypyrrole nanocomposite electrocatalyst for oxygen reduction in an air-breathing direct ethanol fuel cell. The S, N-codoped carbon nanotube catalyst achieved a power density of 20 mW/cm^2 at 0.6 V, exceeding the commercial Pt/C benchmark of 9 mW/cm^2. Finally, this cluster addresses sophisticated current-collector designs, flexible fuel-cell designs, and nano-conducting materials to enhance portability, lightweight, and mechanical flexibility in air-breathing fuel cells. This group of papers clearly indicates that membranes made from CNTs, flexible current collectors supported by polymers, multilayer coatings of metals, and the innovative use of a catalyst/current collector combination would improve conductivity, flexibility, and energy density for future-generation portable PEFCs. Yet the major engineering problem that remains unsolved is sustaining high electrical conductivity, low contact resistance, and electrochemical stability under cyclic loading and severe environmental conditions without compromising its lightweight and flexibility.

3.4.10. Time-dependent performance characteristics of portable PEFCs

Cluster 10 comprises terms including MEA, PEFC, performance characteristics, portable, and time-dependent. This cluster captures a focused, practically important research theme centred on understanding how portable PEFC performance evolves over time during operation. The perspective introduced by "time-dependent" distinguishes this cluster from studies of steady-state performance. It places research squarely within the domain of dynamic characterisation and durability assessment across multiple timescales. The MEA is the heart of the PEFC, comprising the proton exchange membrane sandwiched between two catalyst-coated electrodes. Its properties are not static but evolve continuously during operation due to hydration cycles, catalyst degradation, and ionomer redistribution. Sohn et al. [168] investigated time-dependent performance characteristics in an 80 W air-breathing PEFC system through parametric studies. The reactant humidity and fan-blowing flow rate were examined as key operating variables affecting system performance. Cell voltage distributions were examined over time under various operating conditions to determine the optimal operating strategy. This early study established that time-dependent voltage evolution is strongly linked to humidification management in air-breathing configurations. Mirrashid et al. [169] addressed the time-dependent challenge from a control-systems perspective in air-breathing PEFC systems. The nonlinear, time-dependent characteristics of PEFCs, including complex electrochemical, thermal, and fluid-mechanical phenomena, make controller design challenging. A variable-gain super-twisting algorithm was proposed to regulate the oxygen excess ratio and prevent stack oxygen starvation. The controller maintained an optimal net power output despite simultaneous external disturbances and model uncertainties. Fernández-Moreno et al. [77] constructed and operated a complete portable hydrogen-powered single-cell air-breathing PEFC system. The system performed DC-DC conversion from a low cell voltage of 0.5–0.8 V up to 3.3 V for electronic

applications. System components, including a cathode fan, anodic purging valve, supercapacitors, and a display, were integrated and analysed. The cathode fan and anodic purging helped remove excess water from electrodes, leading to steadier time-dependent cell response. However, these components introduced additional parasitic power consumption, reducing the net system output. The power system delivered >1 W of DC power continuously for 20 h using 1 g of hydrogen. In the work of Chaparro et al. [126], the authors studied the time-dependent water-accumulation behaviour in a passive PEFC using thermal-neutron radiography. Liquid water build-up was imaged over time in both vertical and horizontal orientations during operation. In the vertical position, natural convection and gravity continuously assisted the removal of water, yielding steady power densities above 100 mW/cm^2. In the horizontal position, reduced natural force effectiveness resulted in a 17% decrease in peak power density over the course of operation. Anode flooding caused cell failure after only 5 mg/cm^2 of water production when the cathode faced upward.

Other studies, such as Shen et al. [170], have developed a portable PEFC system using a hydrogenable polyoxometalate solution as a hydrogen carrier. The system operated at atmospheric temperature and pressure without pressurised hydrogen storage. A proof-of-concept air-breathing PEFC achieved good cell performance and stability over time with three different polyoxometalate types. A peak power density of 51 mW/cm^2 was achieved, supporting the practical viability of the polyoxometalate-based portable system. Baek et al. [18] examined time-dependent performance under repeated and sustained mechanical deformation in a foldable air-breathing fuel cell. The foldable design operated in flat, bent, and folded positions without requiring an external air supply system. A peak power density of 36.6 mW/cm^2 was achieved in the folded position during performance evaluation. After 300 cycles of repeated folding, performance was maintained with negligible degradation, confirming excellent cyclic durability. After 24 h of sustained folding, only a slight decrease in performance was observed. Kim et al. [99] investigated ionomer optimisation for self-humidifying air--breathing PEFCs operating under passive conditions, driven solely by natural convection. The study compared the stability of ionomer binders across different equivalent weights and side-chain lengths over time. Ionomer selection was found to have significant time-dependent implications for the stability and durability of passive-type cells. The passive-type cell demonstrated strong performance, making it suitable for compact mobile power sources, including drone applications. Abdulrahman et al. [116] conducted a comparative numerical and Taguchi analysis of the performance characteristics of air-breathing and conventional PEFCs. Cathode GDL thickness was identified as the most influential design parameter affecting performance over time. Air-breathing PEFCs achieved greater sensitivity to cathode GDL porosity than conventional PEFCs due to higher diffusion limitations.

Cluster 10 focuses on the time-dependent performance, dynamic stability, and durable operation of portable PEM fuel cell systems, particularly regarding MEA characteristics, humidification management, water buildup, and transient operation. These studies indicate that optimised system control methods, self-humidified ionomers, and water-management methods could enhance voltage stability, cyclic performance, and power generation from portable fuel cell systems. Nonetheless, the principal technical issue that needs to be addressed is how to ensure consistent electrochemical operation and water management in a dynamic environment of loading, humidity, orientation, and deformability in portable air-breathing PEFCs, while reducing energy loss due to parasitics.

3.4.11. *Thermal management and structural design of air-breathing fuel cells*

Cluster 11 comprises terms including “air breathing”, “fin structure”, “stainless steel bipolar plate”, and “temperature effects”. This cluster focuses on thermal management and structural design of air-breathing fuel cells, with particular attention to fin structures and bipolar plate material selection. Thermal management in air-breathing fuel cells is fundamentally passive, relying on conduction through cell components and natural convection from external surfaces. Excessive temperatures dry out the proton exchange membrane, while insufficient temperatures lead to flooding, making thermal design critically important. Willamson et al. [171] comprehensively investigated the impact of fin structures and temperature variations on air-breathing PEFC performance. The fuel cell body was designed with a large thermal mass relative to its 10 cm^2 active area to prevent self-heating. Three different fin structures were evaluated at four temperatures ranging from room temperature to 50 °C. At higher temperatures, significant enhancement of fuel cell performance was consistently observed across all fin configurations. Schlieren images and calculated Nusselt numbers revealed that fin structures increased convective heat transfer rates and vertical air velocity at elevated temperatures. This enhanced natural convection delivered oxygen to the fuel cell more rapidly, thereby directly improving electrochemical performance. Chun et al. [172] extended this investigation of the fin structure by comparing thin-fin and duct-top-layer arrangements across multiple orientations and temperatures. The authors found that the horizontal configuration recorded the least efficient performance due to water droplet formation on the GDL and to impeded airflow from the large cathode plate. An external heater maintained the fuel cell at the desired test temperatures to minimise fluctuations in self-heating caused by electrochemical reactions. The thin-fin design was deemed a potentially effective top-layer solution due to its simplicity and configurational versatility.

Similarly, the selection of bipolar plate material is a parallel engineering challenge addressed extensively within this cluster. Lee et al. [173] applied metal-coated polycarbonate bipolar plates to simultaneously reduce air-breathing PEFC weight and manufacturing cost. Copper, nickel, and gold layers with a thickness of 50 μm were deposited by electroless plating to eliminate the need for additional current-collector components. Performance degradation due to flooding and membrane drying caused by overheating was observed during operation. A maximum power density of 200 mW/cm^2 was achieved at room temperature with this lightweight plate configuration. Kim and Lee [174] developed lightweight carbon-composite bipolar plates with open-cathode channel configurations for portable air-breathing PEFCs. The design aimed to reduce stack mass by eliminating balance-of-plant components, including fuel pumps, air compressors, and humidifiers. The optimal thickness and stacking sequence of the open-cathode bipolar plate were determined using the maximum strain failure criterion. Kim and Lee [175] further developed an anode bipolar plate and membrane assembly unit bonded with silicone adhesive to address hydrogen leakage. The bonding strength between the Nafion membrane and the carbon fibre/epoxy composite bipolar plate was measured and characterised. The unit cell performance was verified under operating conditions without humidity or temperature control, confirming the design's robustness. Ul Hassan et al. [176] pursued an alternative, low-cost approach using expanded graphite-polymer composite bipolar plates for air-breathing PEFCs. Cutting moulds were designed to pattern fuel-flow channels on thin expanded graphite sheets 0.6 mm thick. Three sheets with flow-channel textures were bonded using commercial conductive epoxy to form the final bipolar plate. The final product achieved a density of 1.79 g/cm^3, with a 20 cm^2 active area plate weighing only 11.38 g. Manufacturing cost was estimated at 7.77 $/kWe with a fabrication time of only two minutes per plate. A total resistance of 22.3 milliohms·cm^2 and a flexural strength of 29 MPa were measured for the three-layer configuration. Stainless steel bipolar plates and their temperature-dependent degradation behaviour were examined in detail by Ramalingam et al. [177]. A 1.0 μm TiN coating was applied to SS304 stainless steel and compared against graphite bipolar plates under long-term test conditions. Tests were conducted at 45 °C and 100% relative humidity, with polarisation curves and EIS recorded every 24 h. TiN-coated SS304 achieved lower resistance, higher hydrophobicity, and greater corrosion resistance than uncoated SS304. Initial fuel cell

performance improved by 147% with a TiN coating, alongside a reduced rate of ohmic resistance increase during operation. After 3000 h of simulated operation, significant changes in resistance were observed, and SEM/EDX analysis revealed coating deterioration and substrate exposure. Cluster 11 addresses issues related to passive thermal management and lightweighting of structures in air-breathing PEFCs, with a specific focus on enhancements through finned convective cooling, materials of construction for bipolar plates, and the effects of temperature on their performance. It is clear from the analysis presented in this cluster that optimal fin design, lightweight bipolar plates, and corrosion-resistant materials enhance fuel cell performance. Nonetheless, the main outstanding engineering constraint is how to realise both long-lasting thermal management and structural resilience in the same lightweight, air-breathing designs, especially during continuous use, when the problems of overheating, oversaturation, corrosion, and coating damage interact to compromise system reliability.

3.4.12. Computational fluid dynamics modelling of ambient air-breathing and tubular fuel cell geometries

Cluster 12 terms include “ambient air-breathing”, “fuel cell modelling”, “tubular”, and “CFD”. This cluster represents a specialised research niche at the intersection of computational fluid dynamics modelling (CFD) and unconventional fuel cell geometries. The cluster is conceptually distinct due to its simultaneous focus on a specific modelling methodology and non-planar cell architectures. CFD is a powerful simulation approach that solves the governing equations of fluid flow, heat transfer, and species transport on discretised computational grids. Sadiq Al-Baghdadi [178] established an early and comprehensive CFD modelling framework for ambient-air-breathing PEM fuel cells. Full three-dimensional, multiphase, non-isothermal computational fluid dynamics models were simultaneously developed for both planar air-breathing and airflow-channel PEM fuel cells. These comprehensive models accounted for convective and diffusive heat and mass transfer, electrode kinetics, and mechanisms of water transport and phase change. The models were shown to capture many interacting, complex electrochemical and transport phenomena that cannot be studied experimentally. This foundational work demonstrated that high-reliability CFD could illuminate fundamental physical understanding of the behaviour of ambient air-breathing fuel cells. Al-Baghdadi [179] subsequently extended this CFD modelling approach to a novel tubular-shaped ambient-air-breathing PEM microfuel cell. The tubular geometry was designed to permit optimum air access to the cathode without pumps, fans, or similar active devices. A three-dimensional, multiphase, non-isothermal CFD model captured convective and diffusive heat and mass transfer, as well as electrode kinetics, within the tubular geometry. The tubular configuration is a significant departure from conventional planar architectures, offering potential structural and convection advantages for portable applications. Kumar and Kolar [127] employed 3D non-isothermal CFD modelling to systematically compare ducted and ribbed cathode designs. The ribbed cathode achieved a peak power density of 232 mW/cm^2 and a limiting current density of 690 mA/cm^2, both substantially higher than those of the ducted cathode. Cell orientation was found to affect performance, with vertical orientation yielding the best results for both cathode designs. Kumar and Kolar [128] further applied 3D CFD to investigate the effects of cathode channel dimensions on the performance of air-breathing PEFCs. Channel width, depth, and cell height were varied systematically across multiple configurations to identify performance trends. Maximum power density of 240 mW/cm^2 was achieved with a channel width of 4 mm and a channel depth of 6 mm. At channel depths below 2 mm, buoyancy-induced flow resistance was the primary performance-limiting mechanism. Above a depth of 2 mm, GDL diffusion resistance became the dominant limiting factor at low current densities.

Other similar studies, including Carcadea et al. [180], conducted numerical CFD investigations of a 44 cm^2 air-breathing PEFC using ANSYS Multiphysics with the Fuel Cell Module. Optimal performance was identified at cathode channel dimensions of 1.5 mm × 1.5 mm through parametric analysis. At operating voltages above 0.75 V, channel width and depth showed only a small influence on performance. Xuan et al. [181] applied integrated CFD and electrochemical kinetics modelling to air-breathing membraneless laminar flow-based fuel cells. Under typical operating conditions, the free-breathing mode was found to provide sufficient oxygen to the electrode's reactive surface. Air breathing was therefore not identified as a limiting factor to cell performance under those conditions. However, oxygen starvation was predicted to become a major limiting factor as the anode was enhanced to achieve higher current density. Sun et al. [155] developed a three-dimensional multiphase CFD model incorporating GDL compression effects and anisotropic conductivity. The optimal open-area ratio for maximum ab-PEFC performance was found to vary with the GDL compression rate. Al-Anazi et al. [182] applied a validated three-dimensional five-cell air-breathing stack model in ANSYS to Saudi Arabia's harsh ambient conditions. Stack performance peaked during the summer months, and hydrogen humidification improved output performance by 40%. Pacheco et al. [122] used CFD to analyse the current collector position in a novel millimetre-sized ab-PEFC design, achieving a power density of 110 W/L. Alziayree et al. [183] introduced a novel radial air-breathing PEM fuel cell developed using 3D multiphase non-isothermal CFD. The radial design accommodated a much larger cathode active area than planar and tubular configurations. At 0.4 A/cm^2, the radial AB fuel cell delivered 27.9% and 10% more power than planar and tubular designs, respectively. Critically, the radial stack design ensured adequate and uniform air supply to all cells, overcoming a fundamental challenge of conventional air-breathing stack configurations. Cluster 12 involves CFD modelling of ambient-air-breathing PEFCs and unique geometries such as tubular and radial designs, with close attention to coupled transport effects, flow distribution, and non-isothermal multiphase operation. From the studies reviewed in this section, it is clear that 3D CFD models can be employed to predict the intricate interplay among heat transfer, species transport, water handling, and electrochemical reactions, while simultaneously optimising novel geometries and stack designs. The main unresolved engineering limitation is the difficulty of translating CFD-optimized unconventional fuel cell designs into experimentally scalable systems that ensure uniform reactant distribution and stable multiphase transport under real-world conditions.

3.5. Conceptual structure

This section presents the conceptual structure analysis, comprising thematic mapping, thematic evolution, and factorial analysis of keywords. It allows a dynamic understanding of how research themes have emerged, matured, or declined over the study period.

3.5.1. Keywords thematic mapping

Fig. 14 maps the author’s keywords into thematic themes. The horizontal axis (Relevance degree/Centrality) measures the degree of interaction between themes, indicating how strongly a research theme connects with other themes in the network. The vertical axis (Development degree/Density) indicates the internal strength and cohesion of each theme, reflecting the intensity of links among keywords within the cluster and the level of development and maturity of that specific research area. The motor themes quadrant (upper right) contains the most critical and well-developed research areas driving air-breathing PEFCs research. Terms such as PEFC, air-breathing, and DMFC dominate this quadrant. The implication is that these topics are both well-developed internally and strongly connected to other research themes across the discipline. This positioning emphasises their role as mature, central research areas with substantial output that shapes the research domain. Similarly, the niche themes quadrant (upper left) includes topics such as piezoelectric, PZT-PEFC, current generation, passive-type

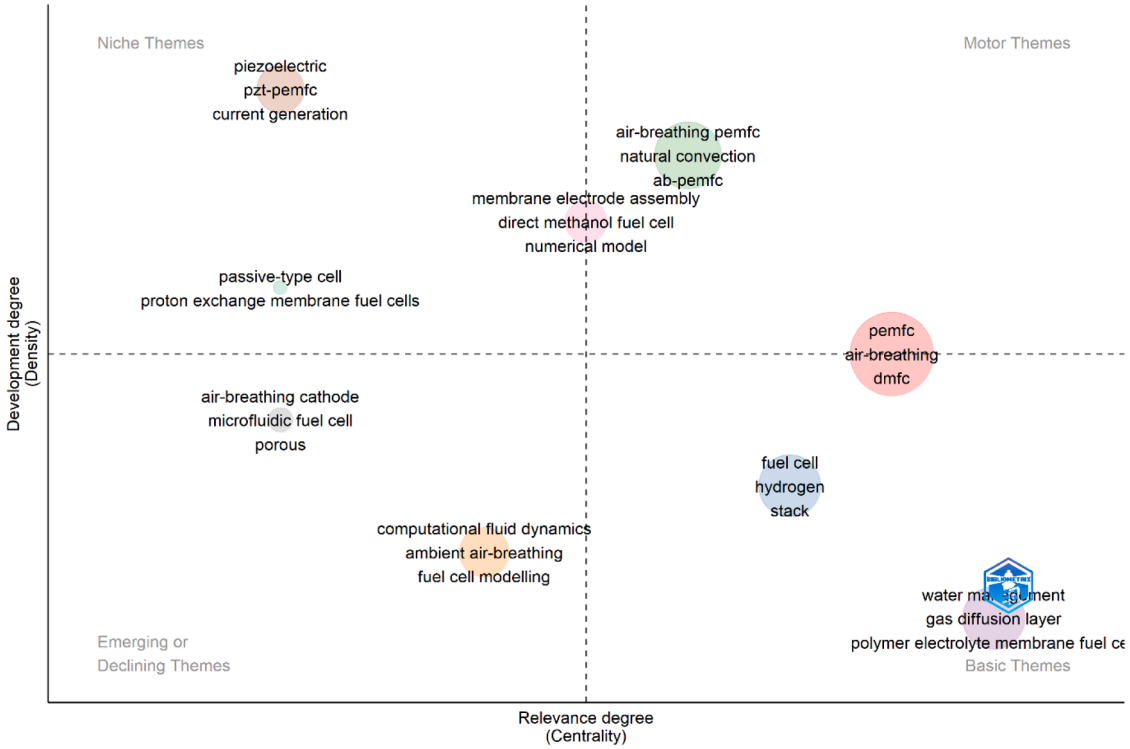


Fig. 14. Thematic mapping of author keywords in air-breathing PEFC research.

cell, and proton exchange membrane fuel cells. These themes show high internal development and density but maintain low centrality. These terms represent well-developed research areas within specialised domains that operate somewhat independently from the mainstream research network. These niche areas demonstrate focused expertise and internal coherence but have limited cross-disciplinary integration with other research streams. Likewise, the basic themes quadrant (lower right) comprises fundamental concepts central to the field but with lower internal density. This quadrant contains essential topics including air-breathing, DMFC, PEFC, fuel cell, hydrogen, and stack. Terms such as water management, gas diffusion layer, and polymer electrolyte membrane fuel cell in this quadrant indicate these are transversal themes that

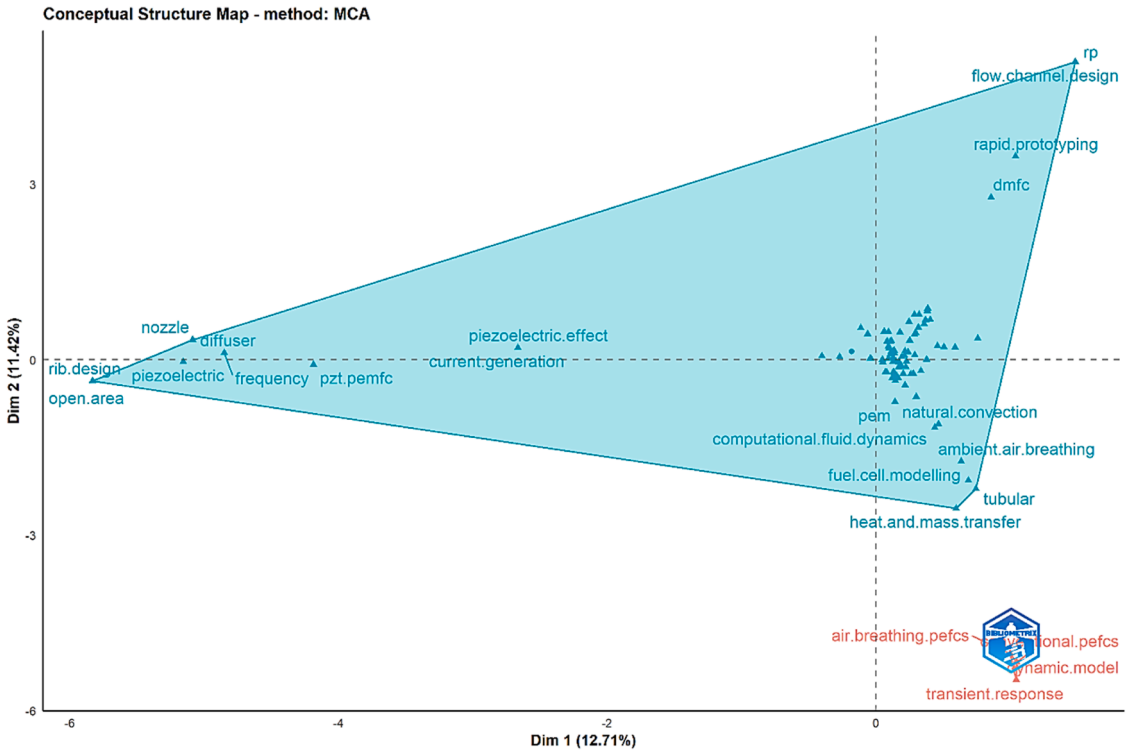


Fig. 15. Factorial analysis of keywords showing thematic clusters in air-breathing PEFC research.

serve as foundational building blocks connecting various research streams. Despite their high relevance and centrality to the field, these topics show relatively lower internal development compared to motor themes, suggesting they represent cross-cutting concerns that permeate multiple research directions rather than forming tightly integrated research clusters.

Furthermore, the emerging or declining themes quadrant (lower left) contains topics with both low centrality and low density, including air-breathing cathode, microfluidic fuel cell, porous, computational fluid dynamics, ambient air-breathing, and fuel cell modelling. These themes occupy marginal positions within the research domain. They represent either emerging research directions that have not yet achieved significant integration and development or declining topics that have lost relevance within the field. The emergence of computational fluid dynamics and fuel cell modelling in this quadrant is noteworthy, as these methodological approaches appear underdeveloped despite their potential to advance fundamental understanding.

3.5.2. Keywords factorial analysis

The conceptual structure map presented in Fig. 15 employs multiple correspondence analyses to identify thematic clusters and conceptual relationships within the literature on air-breathing PEFCs. The two axes represent the primary dimensions of variation in the dataset. Dimension 1 (horizontal) accounts for 12.71% of the total variance, and Dimension 2 (vertical) accounts for 11.42%. These two dimensions capture nearly 24.13% of the total variance in keyword associations, a figure typical of complex, multidisciplinary research fields where numerous distinct yet interconnected themes coexist.

It is worth noting that the spatial positioning of keywords in this figure is not arbitrary; terms that appear closer together indicate stronger co-occurrence in the literature. This suggests that these terms are frequently studied together, represent related concepts, or are associated with similar research methodologies. The colour coding shows two thematic clusters that represent fundamentally different research orientations within the field. The cyan or turquoise cluster, which dominates the upper portion and extends from the left to the right side of the figure, comprises engineering-focused and design-oriented keywords including flow channel design, rapid prototyping, DMFC, computational fluid dynamics, heat and mass transfer, natural convection, ambient air breathing, fuel cell modelling, and various geometric design parameters such as nozzle, diffuser, rib design, and open area. This cluster captures the engineering and optimisation domain of air-breathing PEFC research, where researchers focus on physical design, fluid mechanics, thermal management, and computational modelling to improve system performance. The presence of terms such as piezoelectric effect, frequency, and PZT PEFC within this cluster indicates that active control strategies and advanced actuation mechanisms are considered integral to the engineering optimisation approach.

Conversely, the pink cluster, positioned primarily in the lower right quadrant, contains a more focused set of keywords, including air-breathing PEFCs, conventional PEFCs, dynamic model, and transient response. This cluster shows a more theoretical, comparative research orientation focused on system-level behaviour, dynamic characteristics, and the fundamental differences between air-breathing and conventional fuel cell configurations. The separation between engineering-focused and system-modelling research suggests limited integration between these approaches, potentially offering an opportunity for future research that combines physical design with advanced dynamic modelling. Furthermore, the positioning of certain keywords, such as DMFC and tubular, suggests they represent niche or specialised research areas that may be underexplored compared with planar air-breathing PEFC configurations. The dominance of the engineering cluster in terms of spatial extent and keyword density indicates that the field remains primarily focused on solving practical design and optimisation challenges, which is appropriate for an emerging technology still advancing toward commercial viability. However, this also suggests potential gaps in fundamental understanding of transient behaviour and system dynamics that the smaller modelling cluster attempts to address.

3.5.3. Keywords: thematic evolution from 2000 to 2025

The thematic evolution presented in Fig. 16 illustrates the progression of research keywords and conceptual frameworks across four periods. The vertical coloured boxes represent the research themes, and their heights correspond to each theme's research status. The connecting flow lines indicate the continuity, divergence, or convergence of research topics across successive time periods. The results show that an initial period (2000–2008) was characterised by significant thematic diversity. For example, the dominant theme during this era was air-breathing technology itself, which occupies substantial vertical space on the left side of the figure. This applies to the early stage of air-breathing PEFC technology, where fundamental principles and proof-of-concept studies dominated the literature. Notably, the presence of "pzt-PEMFC" at the top of this column indicates early exploration of novel actuation mechanisms for water management and system optimisation. Several other themes emerged during this foundational period, including conventional fuel cells, proton exchange membrane fuel cells, and air-breathing PEFC variants. The parallel development of these themes suggests that researchers were exploring multiple technological pathways and differentiating air-breathing configurations from traditional closed-cathode systems. The relative size of individual theme boxes during this period, compared with later periods, indicates that the research community was still split and exploring various directions. Similarly, the second period (2009–2013) shows thematic consolidation and the emergence of critical technical challenges. Most notably, "piezoelectric" appears as a notable theme, connected by flows from the air-breathing themes of the previous period. This evolution suggests that researchers identified active water management as a critical bottleneck, leading to increased investigation of piezoelectric actuators for membrane humidification, droplet removal, and enhanced mass transport. The emergence of "water management" as an independent theme during this period is significant. For example, the shift from thick connecting flows across various air-breathing configurations during 2000–2008 to water management during 2009–2013 indicates a consensus within the research community that moisture control is a technical obstacle to the commercialisation of air-breathing PEFC. This is logical because air-breathing cathodes, though they eliminate the need for bulky air-supply systems, still face challenges in maintaining optimal membrane hydration and preventing cathode flooding. In addition, the continued focus on "proton exchange membrane fuel cell" and the emergence of "direct methanol fuel cell" themes suggest that researchers were drawing parallels among different fuel cell configurations and potentially transferring knowledge from more mature technologies to air-breathing systems.

Likewise, the third period (2014–2018) shows a transition toward application-oriented research themes. The figure shows "fuel cell" and "PEM fuel cell" as dominant themes, with substantial connecting flows from the previous period's water management and piezoelectric research. This evolution indicates that solutions to fundamental challenges have enabled researchers to focus on system-level integration and practical implementations. The continued presence of "water management" demonstrates that this challenge remained central to the field, though the research shifted from identifying the problem to developing solutions. The appearance of "proton exchange membrane" as a distinct theme, separate from complete fuel cell systems, suggests increased attention to materials development, membrane electrode assembly optimisation, and component-level innovations. Notably, "air-breathing" persists through all these transformations, indicating that this core concept remained relevant even as research questions evolved. The multiple parallel flows from various 2009–2013 themes converging into 2014–2018 topics suggest a developing field where different research streams began to integrate their findings. Furthermore, the most recent period (2019–2025) shows an interesting trend of both integration and

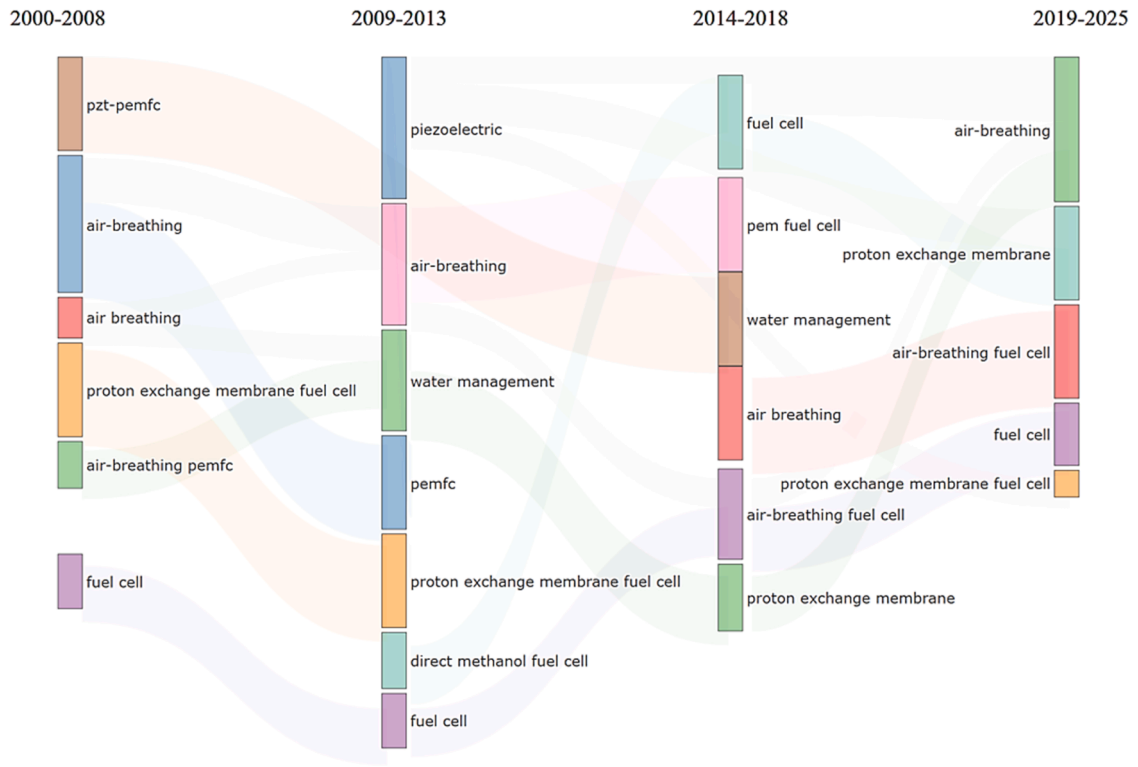


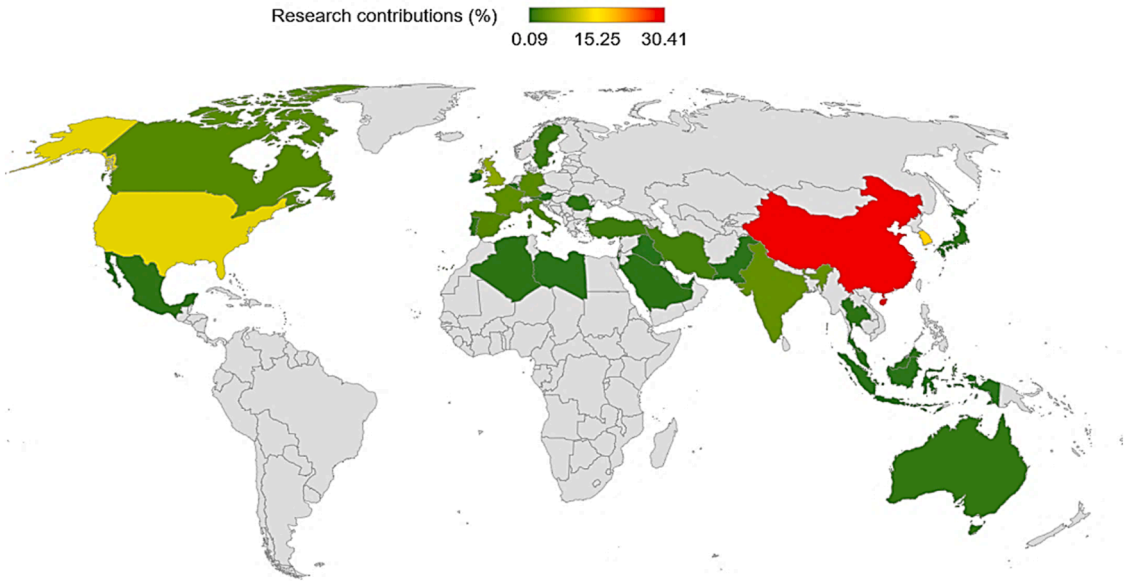
Fig. 16. Thematic evolution of research topics in air-breathing PEFC studies.

renewed diversification. "Air-breathing" reemerges as the dominant theme, presumably incorporating decades of accumulated knowledge in water management, materials optimisation, and system design. The large box size for air-breathing in this period, compared to earlier eras, suggests not just research volume but also the breadth of subtopics now encompassed under this umbrella term. The presence of "proton exchange membrane" as a substantial independent theme indicates continued innovation in materials. The persistence of "fuel cell" and "air-breathing fuel cell" as different themes suggests that the field now comprises both fundamental research and specialised applications of the air-breathing concept. It is worth noting that the figure shows multiple convergence points where themes from 2014–2018 flow into the 2019–2025 period, indicating knowledge synthesis and the field's growth toward practical applications. The fact that themes such as water management feed into contemporary air-breathing research suggests that earlier challenges have been sufficiently addressed, enabling a focus on system-level performance and deployment.

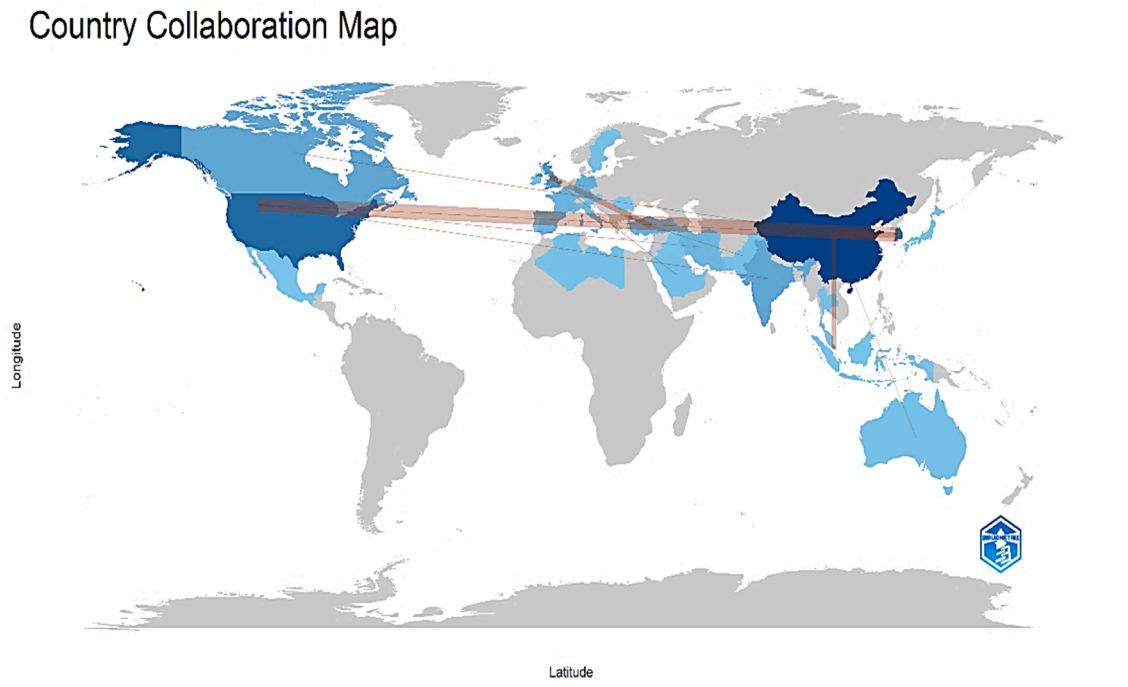
3.6. Country research contributions and collaboration patterns

Fig. 17(a) shows countries contributing to air-breathing PEFC research worldwide. China (30.41%), South Korea (17.04%), and the United States (12.71%) account for approximately 60% of all research contributions. China's status as the leading contributor in this research domain can be attributed to comprehensive national and provincial policies in China [184] that prioritise fuel cell technology as a pillar of energy security and environmental sustainability. The Chinese government has implemented financial mechanisms, including the National Key Research and Development Program [185], which channels billions of yuan into hydrogen and fuel cell research through competitive grants to universities and research institutes. However, the figure shows a disparity in research contributions, with numerous countries contributing <1% of global output, including Japan (0.38%), Australia (0.75%), and several Middle Eastern nations. Countries with limited contributions face a critical juncture in which strategic investment in air-breathing PEFC research could yield disproportionate returns relative to their investment, especially for nations with specific contextual advantages or urgent energy security needs. Middle Eastern countries such as Saudi Arabia (0.38%), the United Arab Emirates (0.09%), and Iraq (0.09%), despite their currently minimal research contributions, have compelling reasons to substantially expand their air-breathing PEFC research programs. This expansion could form part of economic diversification strategies to reduce oil dependency. They can leverage their substantial financial resources, abundant solar energy for hydrogen production via electrolysis, and strategic geographic positioning along global shipping routes where air-breathing fuel cells could power auxiliary systems in maritime applications.

Similarly, nations in Southeast Asia, including Thailand (0.38%), Indonesia (0.28%), and Malaysia (0.38%), face severe urban air pollution and energy security challenges, making air-breathing PEFC technology mainly relevant for distributed power generation and two-wheeler electrification. African and Latin American nations, represented only by Algeria (0.38%) and Mexico (0.47%), face unique opportunities in which air-breathing PEFC technology could address off-grid electrification needs and provide reliable power for telecommunications infrastructure in remote regions. These nations' limited contributions reflect broader challenges in research funding, human capacity development, and international collaboration networks. These regions would benefit enormously from targeted international development programs, perhaps through mechanisms such as the Green Climate Fund or bilateral technology transfer agreements with leading research nations, that specifically support fuel cell research capacity building through equipment grants, scholar exchange programs, and twinning arrangements between emerging and established research institutions. The broader implication of this research focus is that innovation in air-breathing PEFC technology may be constrained by limited diversity in research perspectives, environmental testing conditions, and application contexts, potentially slowing the technology's adaptation across diverse global markets and use cases. Therefore, countries with minimal current contributions should recognise that strategic investment in this research



(a)



(b)

Fig. 17. (a) Country-level research contributions and (b) international collaboration networks in air-breathing PEFC research.

domain represents not merely scientific advancement but a pathway to participation in the emerging hydrogen economy, development of high-value manufacturing capabilities, and cultivation of expertise that will prove essential as global energy systems transition away from fossil fuels over the coming decades.

The country collaboration map in Fig. 17(b) illustrates the global research network. The colour intensity indicates the degree of research activity and collaboration. Dark blue regions indicate countries with high research output and extensive collaborative partnerships. In contrast, the lighter shades of blue denote moderate research activity, and grey represents minimal or no participation in this research domain. The connecting lines between countries visualise bilateral or multilateral research collaborations, with their thickness potentially reflecting the strength or frequency of partnerships. The results reveal a highly concentrated collaboration network dominated by a few countries. The United States and China dominate as the main research hubs, with the darkest blue colouration, and serve as central nodes with multiple collaboration lines extending to other regions. This implies that these nations not only lead in research productivity but also serve as collaborative bridges within the global network. European countries, including Germany and the United Kingdom, as well as several Asian nations such as South Korea, India, and Southeast Asian countries, demonstrate moderate to strong engagement, forming secondary clusters within the network. Australia's participation is evident through its light blue colouration and collaboration links to the primary hubs. Notably, Africa and South America appear disconnected mainly from

this research network, highlighting significant geographical disparities in participation in fuel cell research. These patterns suggest broader trends in global scientific collaboration, where technological advancement concentrates in economically developed regions with established research infrastructure. The strong US-China collaboration axis, despite geopolitical tensions, emphasises the pragmatic nature of scientific partnerships in emerging energy technologies. However, the limited participation from developing regions raises concerns about knowledge transfer and technological equity in clean energy transitions. The heavily centralised network structure, while facilitating rapid knowledge exchange among leading institutions, may create vulnerabilities and limit diverse perspectives in fuel cell development. Thus, strengthening South-South collaborations and integrating regions with limited collaboration could enhance the robustness and global relevance of air-breathing fuel cell research for diverse environmental and economic applications.

4. Summary of key insights and research outlook

This review has identified several critical insights into the evolution, thematic structure, and geographic distribution of air-breathing PEFC research. The findings show that air-breathing PEFC research is dominated by China (30.41%), South Korea (17.04%), and the United States (12.71%), accounting for about 60% of global research output. However, significant portions of the world, including Africa, Latin America, the Middle East, and Southeast Asia, have limited contributions. This limitation risks constraining the diversity of environmental testing conditions, application scenarios, and engineering design perspectives that are essential to developing truly versatile, globally deployable systems. The geographic and collaboration analysis therefore provides an important contribution. It emphasises that the future development of air-breathing PEFCs should not be considered only as a technological challenge, but also as a question of research capacity, geographical contributions, and international collaboration.

Similarly, the keyword co-occurrence analysis identified 12 thematic clusters comprising thermal and mass transport phenomena, piezoelectric actuation for enhanced water management, flow field and numerical modelling, performance diagnostics, high-temperature operation, miniaturised direct methanol fuel cell design, gas diffusion layer engineering, ambient air quality effects, flexible current collector architectures, time-dependent performance characterisation, thermal management and structural design, and computational fluid dynamics of unconventional geometries. Water management emerged as the most notable technical challenge, with the highest keyword frequency in the word cloud and forming the thematic axis around which numerous subsidiary research themes, including GDL design, cathode open ratio optimisation, membrane material development, and computational modelling, are organised. The thematic evolution analysis shows that research gradually consolidated around water management between 2009 and 2013, then shifted toward broader system-level integration and application-oriented goals in subsequent periods. This temporal perspective further contributes to the study by showing how the field has progressed from addressing individual components and transport limitations toward increasingly integrated system-level objectives.

Likewise, the thematic mapping and factorial analysis identified a structural bifurcation between engineering-focused design research and system-level dynamic modelling. However, CFD and transient modelling approaches is captured in the emerging or declining thematic quadrant and within a smaller conceptual cluster, respectively, despite their demonstrable potential to illuminate complex coupled multi-physics behaviour under passive operating conditions. This bifurcation suggests a prospect for integrating computational and experimental paradigms. The results therefore reveal a central research gap in the field. Although substantial knowledge has been generated on component design and passive transport phenomena, comparatively less attention has been given to integrating these findings within validated dynamic, non-isothermal, and experimentally informed system-level models. Therefore, addressing this gap is essential to move beyond empirical optimisation toward the predictive design of air-breathing PEFC systems. The study's specific objectives, principal findings, and key gaps identified are summarised in Table 3.

Table 3
Summary of objectives, main findings, and identified gaps in air-breathing PEFC research.

Focus of the objective	Main Findings	Identified gaps
Analyse annual publication trends (2000–2025)	Research shows steady but non-linear growth. Early years were marked by exploratory work, followed by peaks linked to a global drive for hydrogen and climate policy. Growth rate indicates sustained interest but fluctuates with external funding and energy-sector dynamics.	Research activity remains vulnerable to policy and funding cycles. Limited evidence of sustained large-scale demonstration projects to stabilise long-term research momentum.
Identify leading countries and international collaboration patterns	The field is highly collaborative at the author level (multi-authored publications dominate). However, the international co-authorship rate remains relatively modest, indicating concentration within national or regional networks.	Limited cross-border collaboration may restrict the diffusion of global knowledge and coordinated technological advancement. Greater international integration is needed to scale and standardise.
Examine dominant and emerging research themes.	Water management, performance optimisation, and thermal-mass transport phenomena dominate. Strong emphasis on cathode design, GDL optimisation, membrane materials, modelling (CFD), and flow-field geometry. Emerging themes include piezoelectric-assisted airflow and rapid prototyping for miniaturised systems.	Persistent flooding-drying trade-off remains unresolved. Limited integration between component-level optimisation and full system reliability. Need for cost-effective catalyst systems under passive conditions.
Evaluate the evolution of the conceptual structure	The field has evolved from foundational electrochemical and modelling studies to integrated system-level optimisation. Increasing focus on high-temperature membranes (e.g., PBI), stack configuration, UAV applications, and environmental sensitivity. Modelling tools are coupled with experimental validation.	Insufficient long-term durability studies under realistic cyclic and environmental conditions. Lack of unified multi-physics optimisation frameworks linking water, heat, and structural design.
Determine key research clusters through keyword co-occurrence analysis.	Major clusters include: (i) thermal and mass transport phenomena, (ii) piezoelectric-enhanced mass transport, (iii) flow-field and MEA design optimisation, (iv) portable fuel cell diagnostics, (v) durability and high-temperature operation, (vi) miniaturised DMFC systems, and (vii) GDL/MPL water management strategies. These clusters emphasise passive oxygen transport and environmental sensitivity as defining characteristics.	Scale-up challenges remain limited, mainly stack-level thermal uniformity and contact resistance management. Net power evaluation in hybrid PZT systems requires further refinement. Standardised benchmarking protocols specific to air-breathing configurations are lacking.

Drawing on these insights, the following research areas are critical priorities for advancing the field of air-breathing PEFCs. These priorities are derived from the gaps identified through the bibliometric, thematic, and conceptual analyses and are intended to provide a forward-looking research agenda for the field. First, the factorial analysis shows limited integration between engineering-focused design research and dynamic modelling approaches, with the two orientations occupying distinct spatial clusters in the conceptual structure map. Therefore, future research should systematically couple transient and non-isothermal computational models with parametric physical design frameworks to accurately capture time-dependent phenomena during load cycling, start-up transients, and ambient-condition fluctuations that steady-state models cannot resolve. Such models should be experimentally validated across multiple operating conditions and should incorporate coupled heat, mass, charge, and water transport processes. This would help establish a stronger connection between computational predictions and experimentally measurable performance, thereby addressing the current separation between modelling and engineering design.

Secondly, most studies in the analysed dataset remain confined to short-term performance evaluations, and the durability cluster, while present, remains limited relative to its commercial importance. Hence, systematic multi-thousand-hour in situ and ex situ degradation studies under varied humidity, temperature, and contaminant exposure conditions are essential to establish failure mechanisms, quantify component-level deterioration rates, and develop predictive lifetime frameworks specifically applicable to passive air-breathing configurations operating without active humidification or thermal control. The lack of standardised long-duration testing protocols is therefore another significant gap. Future studies should establish common durability metrics, operating protocols, degradation indicators, and reporting procedures to enable comparison of results across different laboratories and cell architectures.

Thirdly, the multiphase modelling and ambient air quality cluster highlights the limited impact of real-world atmospheric pollutants on the air-breathing performance of PEFCs. Notable studies mainly tested individual contaminants under controlled laboratory conditions. Future research should therefore investigate complex, multi-contaminant exposure scenarios, including exposure to NOx, CO, particulate matter, and sulphur compounds, under dynamic atmospheric conditions representative of urban, industrial, and military deployment environments. This implies an important transition from laboratory-based performance assessment toward application-relevant environmental testing. Therefore, future studies should quantify the interactions between pollutants, humidity, temperature, oxygen availability, and water management rather than treating these variables independently.

Fourthly, integrated mitigation strategies, such as catalytic air-filtration interlayers or contaminant-resistant GDL surface coatings, should be systematically evaluated. However, their effectiveness should be assessed with potential penalties in pressure drop, mass transport, cost, weight, manufacturability, and long-term durability. This integrated assessment is necessary to determine whether pollutant mitigation strategies can provide practical benefits without compromising the fundamental advantages of passive air-breathing architectures. Fifthly, the flexible current collector cluster demonstrates the feasibility of ultra-thin, rollable, and foldable air-breathing PEFCs for next-generation portable electronics and wearable power systems. However, fundamental questions remain regarding mechanical durability under high-cycle deformation, the optimisation of current collector design for large active areas, and the long-term electrochemical stability of flexible membrane electrode assemblies. Therefore, dedicated research programmes targeting scalable fabrication methods, interfacial contact management under bending, and standardised mechanical-electrochemical durability protocols are essential for transitioning proof-of-concept demonstrations into commercially viable platforms. Future work should also move beyond demonstrations at small active areas and establish whether the favourable electrochemical performance of flexible architectures can be retained during repeated bending, folding, stretching, and practical device integration.

Similarly, the GDL and microporous layer cluster highlights the essence of porous-transport media design in passive systems, where the driving force for reactant delivery is limited to natural convection and diffusion alone. Therefore, research into biologically inspired and sustainably sourced GDL materials, including alveolar-inspired integrated architectures, natural wood-derived carbon sheets, and ultra-thin carbon nanotube-based layers, has yielded promising improvements in power density and water management. Future studies should systematically characterise these novel architectures under realistic passive operating conditions, with emphasis on scalable and cost-effective manufacturing routes, compression-dependent transport behaviour, and compatibility with advanced microporous layer compositions. A key remaining gap is the limited understanding of how these novel porous architectures behave under compression, liquid-water accumulation, temperature gradients, and long-term degradation, which affect their transport properties. Addressing this gap will require combined experimental characterisation and multi-physics modelling at both material and cell scales.

Likewise, the miniaturisation and flow-field cluster, along with computational studies of UAV and high-altitude applications, reveal that air-breathing PEFCs operating at altitude experience severe performance reductions due to diminished oxygen partial pressure, altered humidity, and lower ambient temperatures. Future research specifically targeting performance modelling under varying atmospheric conditions, MEA design optimisation for low-pressure operation, and integrated thermal management strategies for thermally extreme environments is strategically significant but largely underdeveloped, with direct relevance to autonomous aerial systems and high-altitude distributed power applications. Future investigations should therefore incorporate realistic altitude profiles and transient atmospheric conditions rather than relying solely on fixed laboratory pressure and temperature conditions. Such work would improve the transferability of laboratory findings to autonomous aerial systems and other high-altitude applications.

It is worth mentioning that the principal future perspective emerging from this review is that the field should progress from isolated optimisation of individual components toward integrated, durable, environmentally strong, and application-specific air-breathing PEFC systems. The most important gaps are not limited to individual materials or components. They also concern the limited integration of computational and experimental approaches, insufficient evidence of long-term durability, inadequate representation of realistic atmospheric conditions, limited validation of flexible and miniaturised architectures, and substantial geographic disparities in research activity. These gaps can be addressed through coordinated multi-physics modelling and experimentation, standardised long-term testing, realistic multi-contaminant environmental exposure studies, scalable materials and manufacturing research, and stronger international research collaboration. Therefore, addressing these priorities would provide a pathway to translate the substantial body of existing laboratory research into reliable, globally deployable air-breathing PEFC technologies.

5. Conclusion

This study provided a structured, systematic bibliometric review of air-breathing PEFC research from 2000 to 2025. The findings confirm that air-breathing PEFCs have evolved from early exploratory investigations into a specialised but steadily developing research domain within the broader fuel cell domain. Although publication growth has been non-linear, the sustained output and increasing technical sophistication demonstrate continued scientific and engineering relevance. The review reveals that water management remains the central technical constraint in air-breathing systems. The passive oxygen supply mechanism intensifies the balance required between membrane hydration and cathode flooding, directly affecting performance, durability,

and reliability. Consequently, substantial effort has been directed toward gas-diffusion layer engineering, microporous-layer optimisation, catalyst-layer structuring, and cathode open-area design. These studies show that geometric optimisation, hydrophobic-hydrophilic balance, and controlled porosity are decisive for improving oxygen transport and liquid water removal under natural convection conditions. Similarly, thermal management also emerged as a critical factor, mainly for stack-level and high-current-density operation. The coupling of heat and mass transfer limitations in passive systems means that performance cannot be addressed in isolation from temperature distribution and environmental conditions. Advances in numerical modelling, including CFD-based multi-physics simulations, have significantly improved understanding of transient behaviour, altitude effects, and design trade-offs. These tools now play a central role in guiding experimental optimisation and reducing development cycles. Likewise, material innovation is another important implication of the findings. For example, developments in crosslinked membranes, polybenzimidazole-based high-temperature systems, nanocomposite structures, and platinum-free catalysts demonstrate ongoing efforts to enhance durability and reduce cost. In addition, emerging concepts such as piezoelectric-assisted airflow and rapid prototyping of miniature stacks indicate that the field is expanding beyond purely passive designs toward hybrid and application-specific solutions. From a stakeholder perspective, the study underlines the importance of interdisciplinary collaboration. Researchers must integrate electrochemistry, materials science, fluid mechanics, and thermal engineering to address system-level constraints. For industry, the results highlight practical design priorities: robust water-management strategies, lightweight, compact stack architecture, and reliable operation across variable ambient conditions. Policymakers and funding agencies can use these insights to support targeted innovation in portable, off-grid, and UAV-compatible hydrogen power systems. The future of air-breathing PEFCs depends on bridging the gap between laboratory optimisation and the scalable, durable integration of systems. Thus, continued progress in materials engineering, passive flow-field design, and predictive modelling will be essential to achieving broader commercial viability. It should be noted that this study's literature search was limited to the Scopus database. Therefore, future studies could utilise additional databases, such as Web of Science or IEEE Xplore, to compare and validate the findings.

CRediT authorship contribution statement

Flavio Odoi-Yorke: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Ephraim Bonah Agyekum:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **John Eshun Davis:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **Feras Alnaimat:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **Arman Ameen:** Writing – review & editing, Project administration, Formal analysis. **Seepana Praveenkumar:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis. **Farhan Lafta Rashid:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

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